

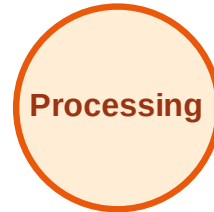
# BFS Traversal

Step 1: Start at A

## Legend



Complete



Processing



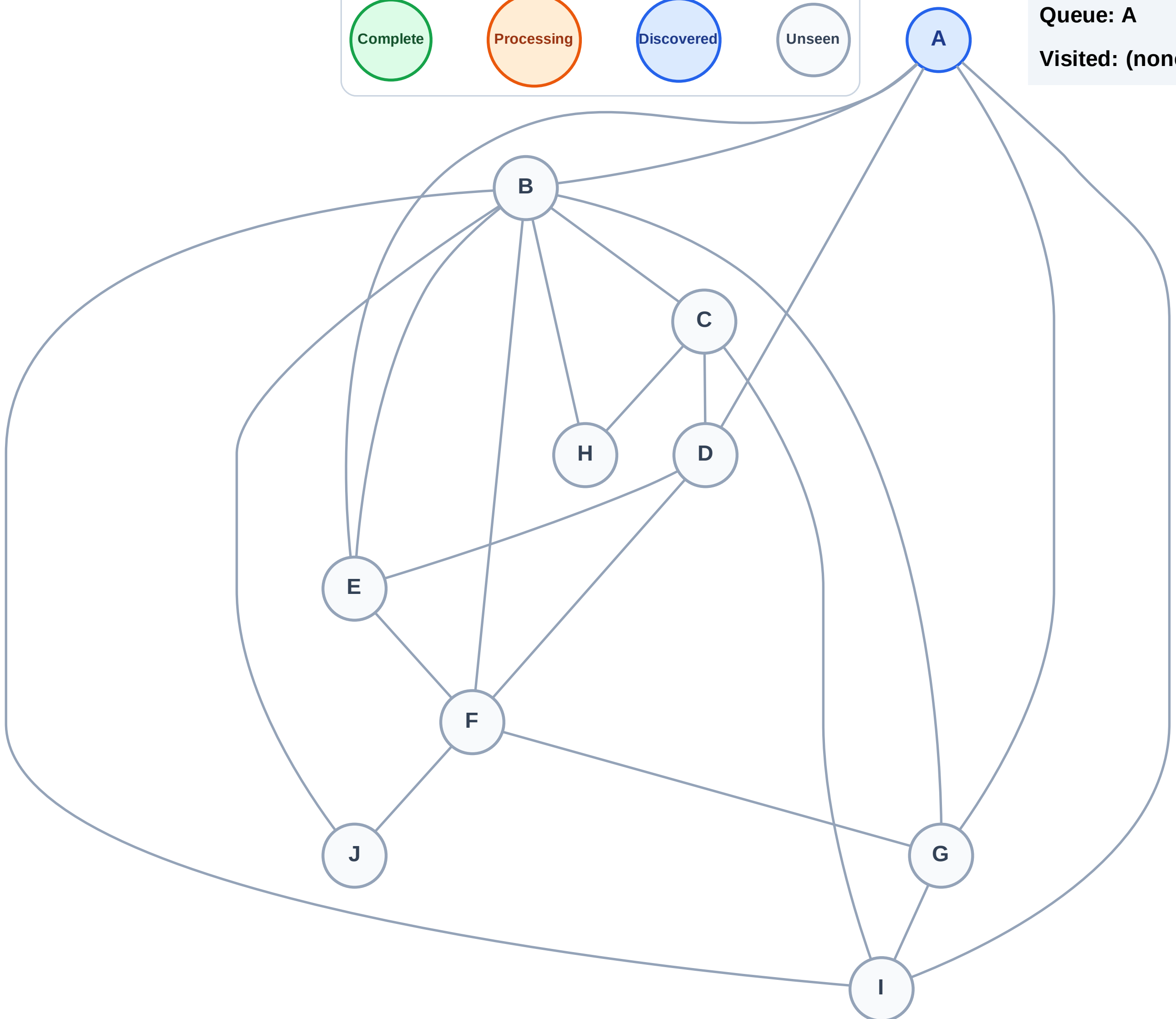
Discovered



Unseen

Queue: A

Visited: (none)



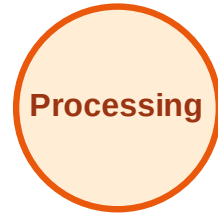
# BFS Traversal

Step 2: Process node A

## Legend



Complete



Processing

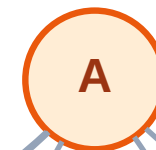


Discovered



Unseen

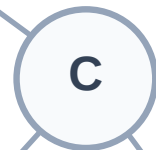
Queue: (empty)  
Visited: (none)



A



B



C



H



D



E



F



J



G

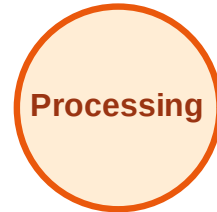


I

# BFS Traversal

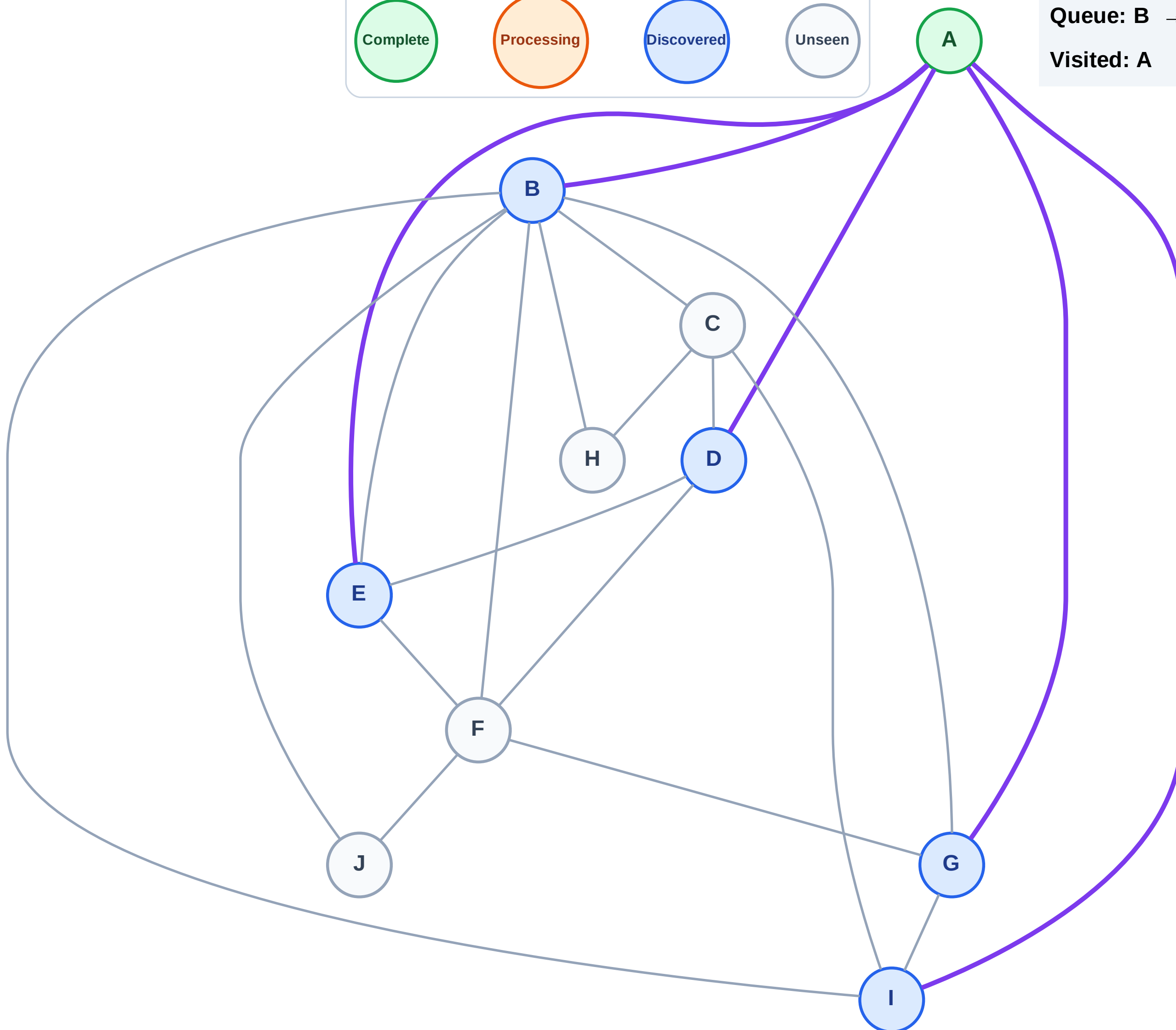
Step 3: Discover B, D, E, G, I from A

## Legend



Queue: B → D → E → G → I

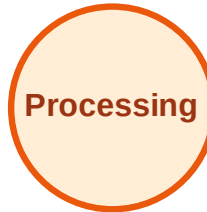
Visited: A



# BFS Traversal

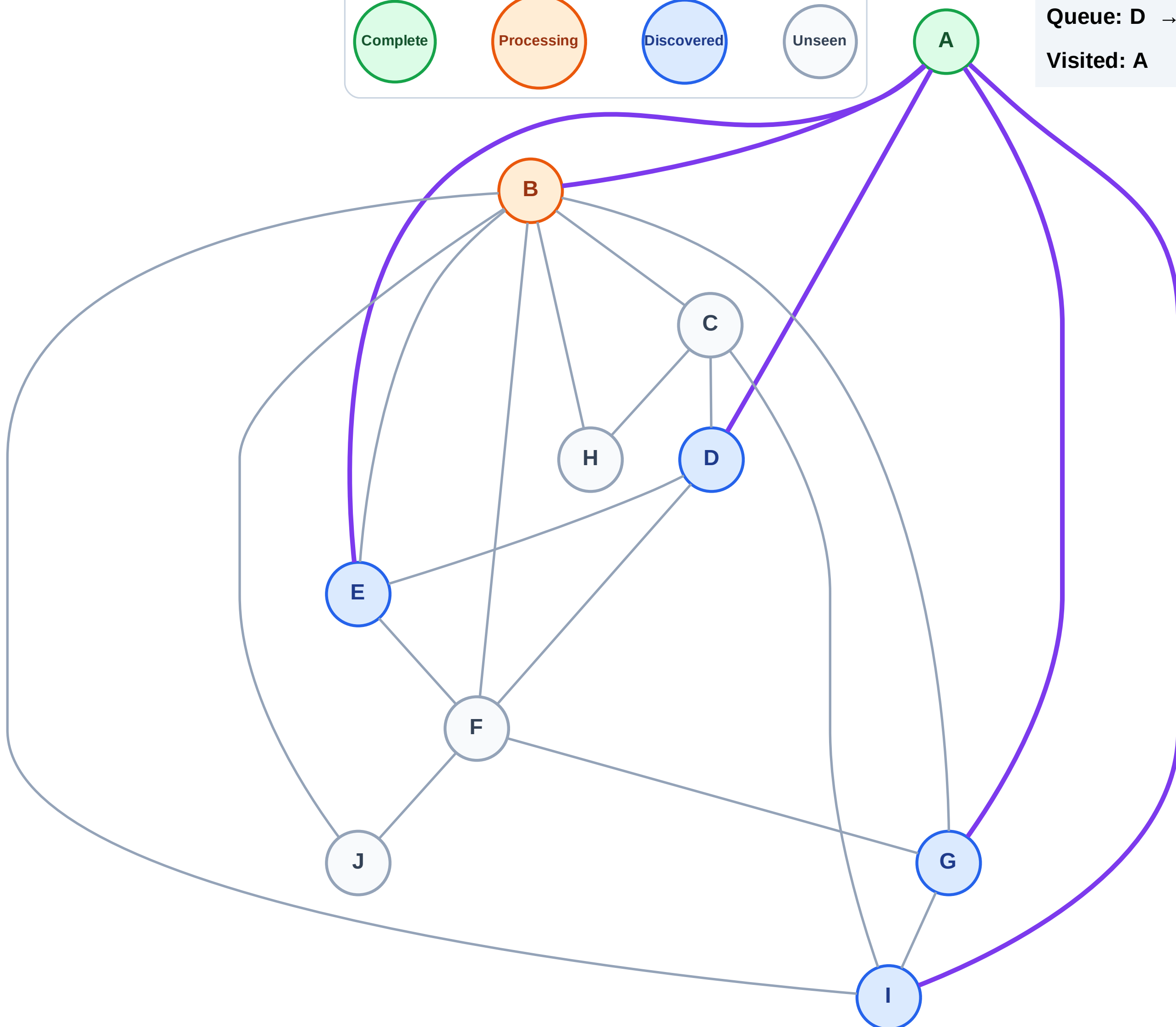
Step 4: Process node B

## Legend



Queue: D → E → G → I

Visited: A



# BFS Traversal

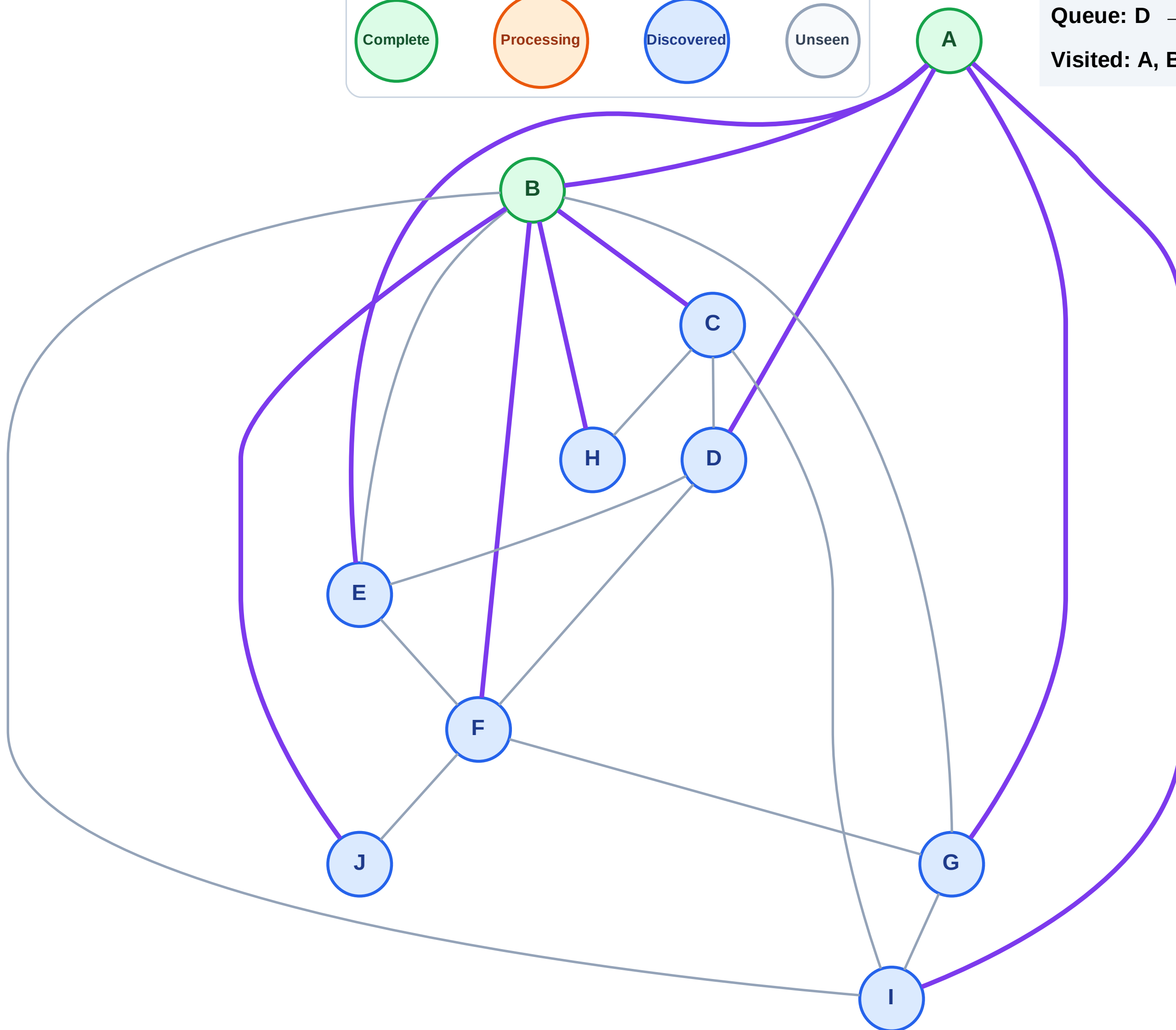
Step 5: Discover C, F, H, J from B

Legend



Queue: D → E → G → I → C → F → H → J

Visited: A, B



# BFS Traversal

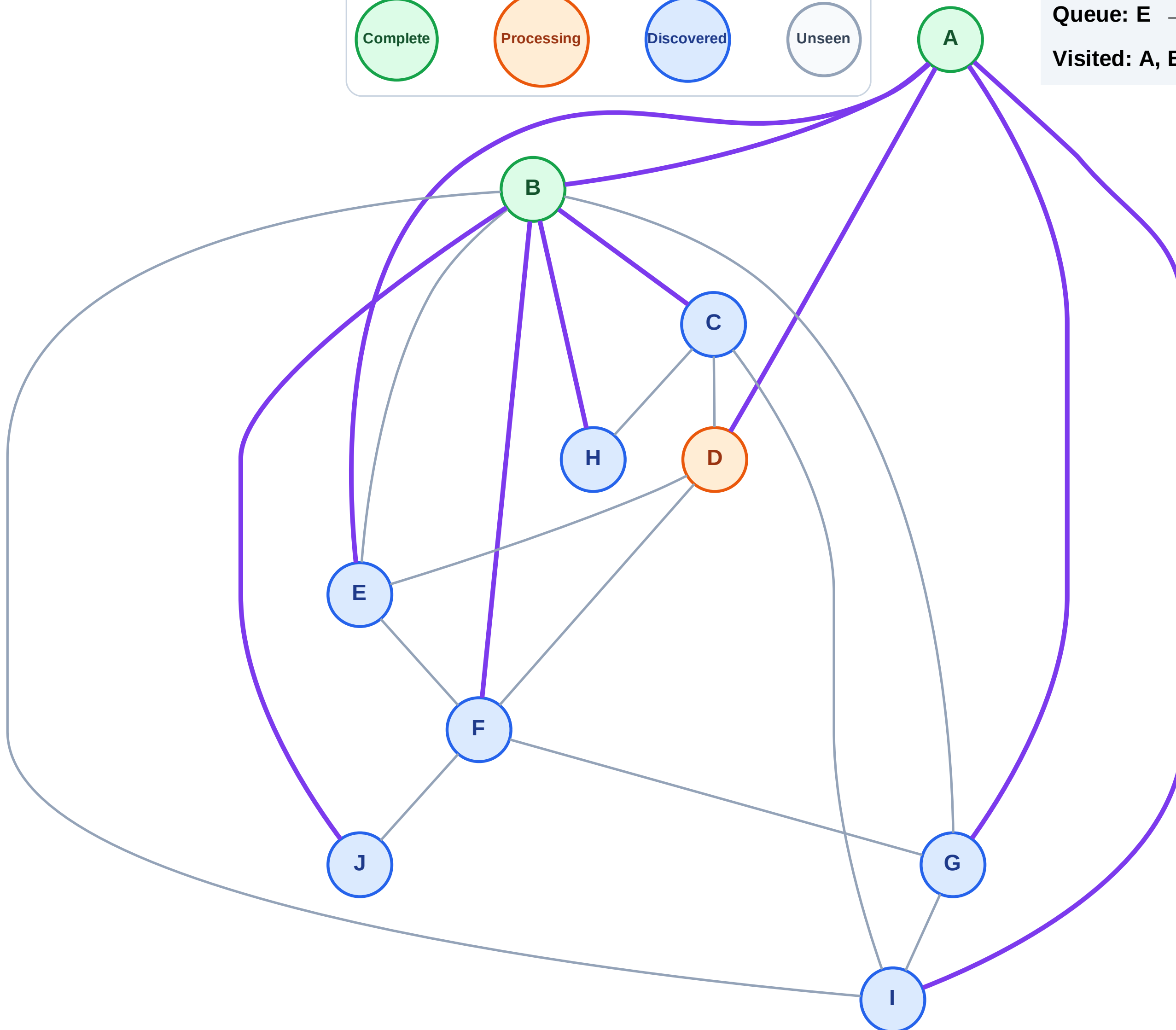
Step 6: Process node D

## Legend



Queue: E → G → I → C → F → H → J

Visited: A, B



# BFS Traversal

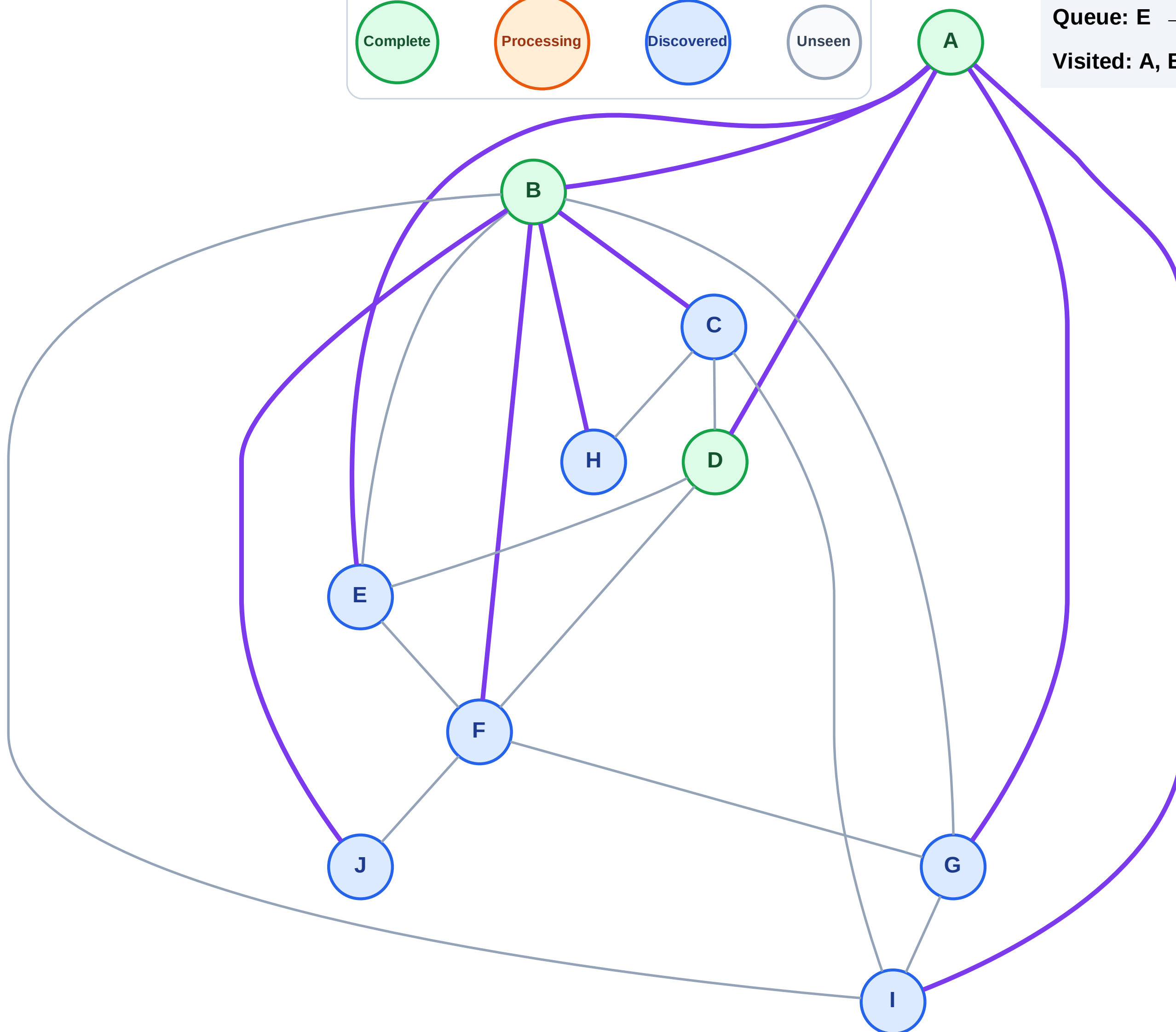
Step 7: No new neighbors from D

## Legend



Queue: E → G → I → C → F → H → J

Visited: A, B, D



# BFS Traversal

Step 8: Process node E

Legend

Complete

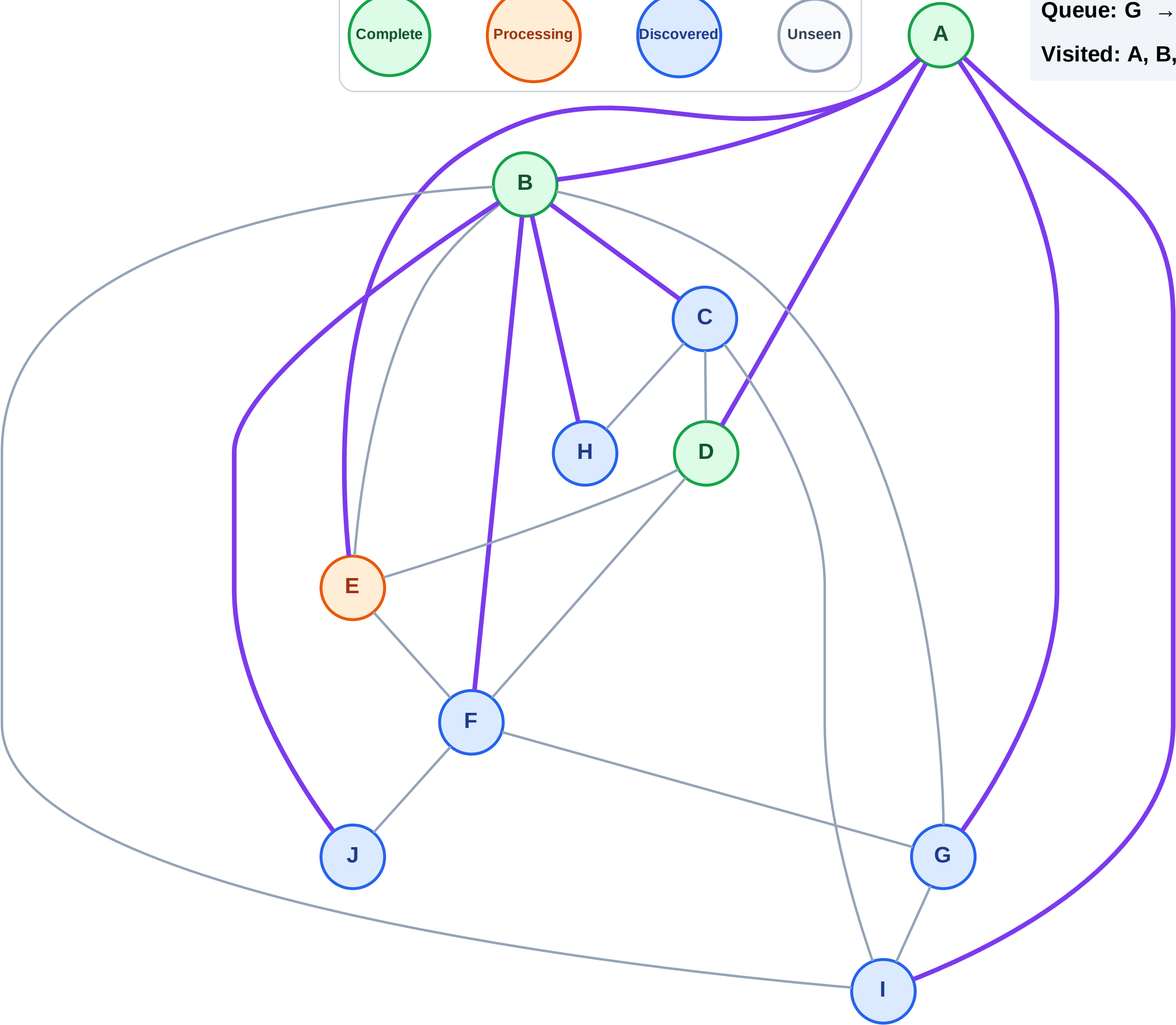
Processing

Discovered

Unseen

Queue: G → I → C → F → H → J

Visited: A, B, D



# BFS Traversal

Step 9: No new neighbors from E

Legend

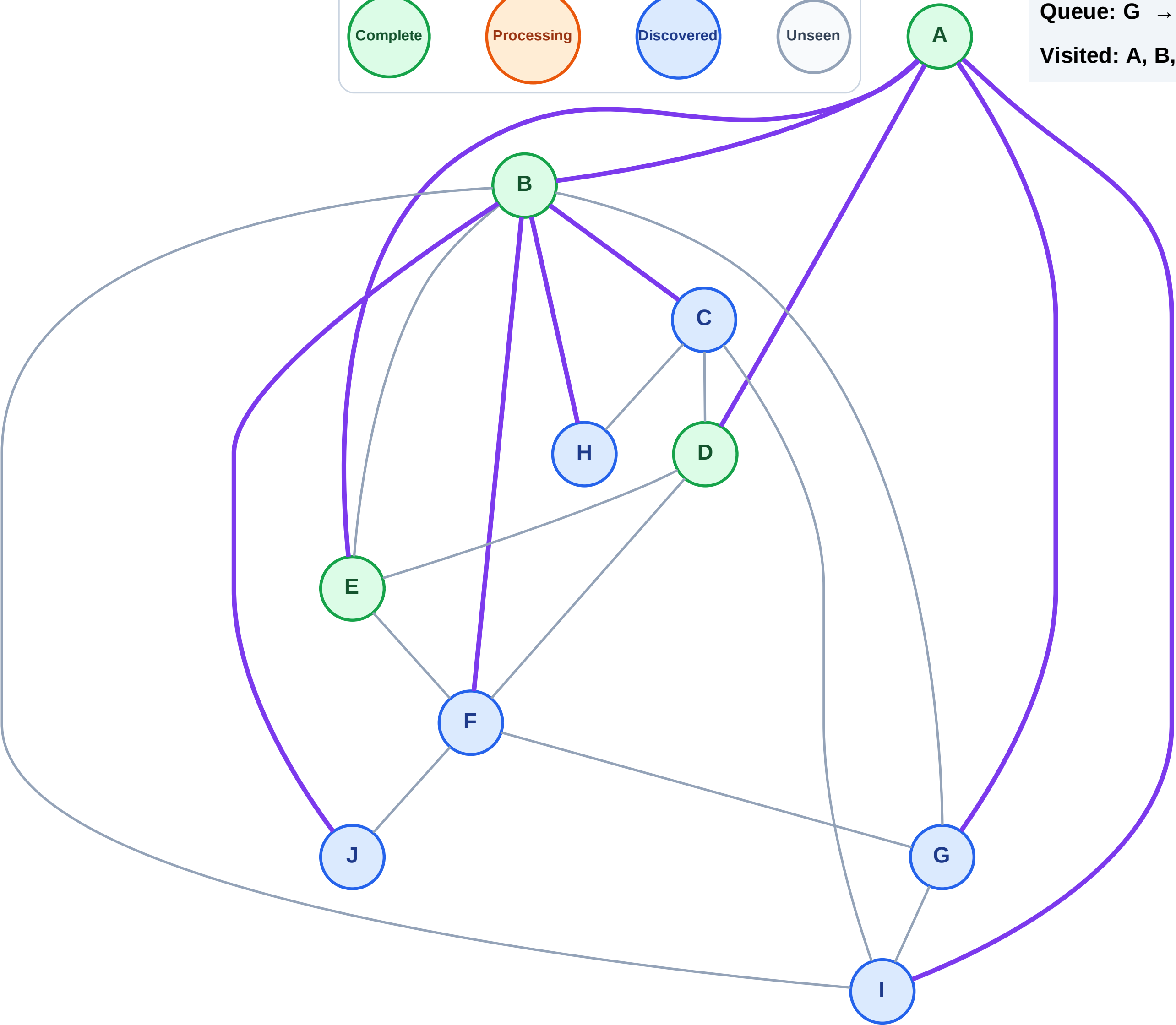
Complete

Processing

Discovered

Unseen

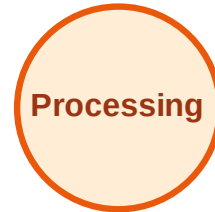
Queue: G → I → C → F → H → J  
Visited: A, B, D, E



# BFS Traversal

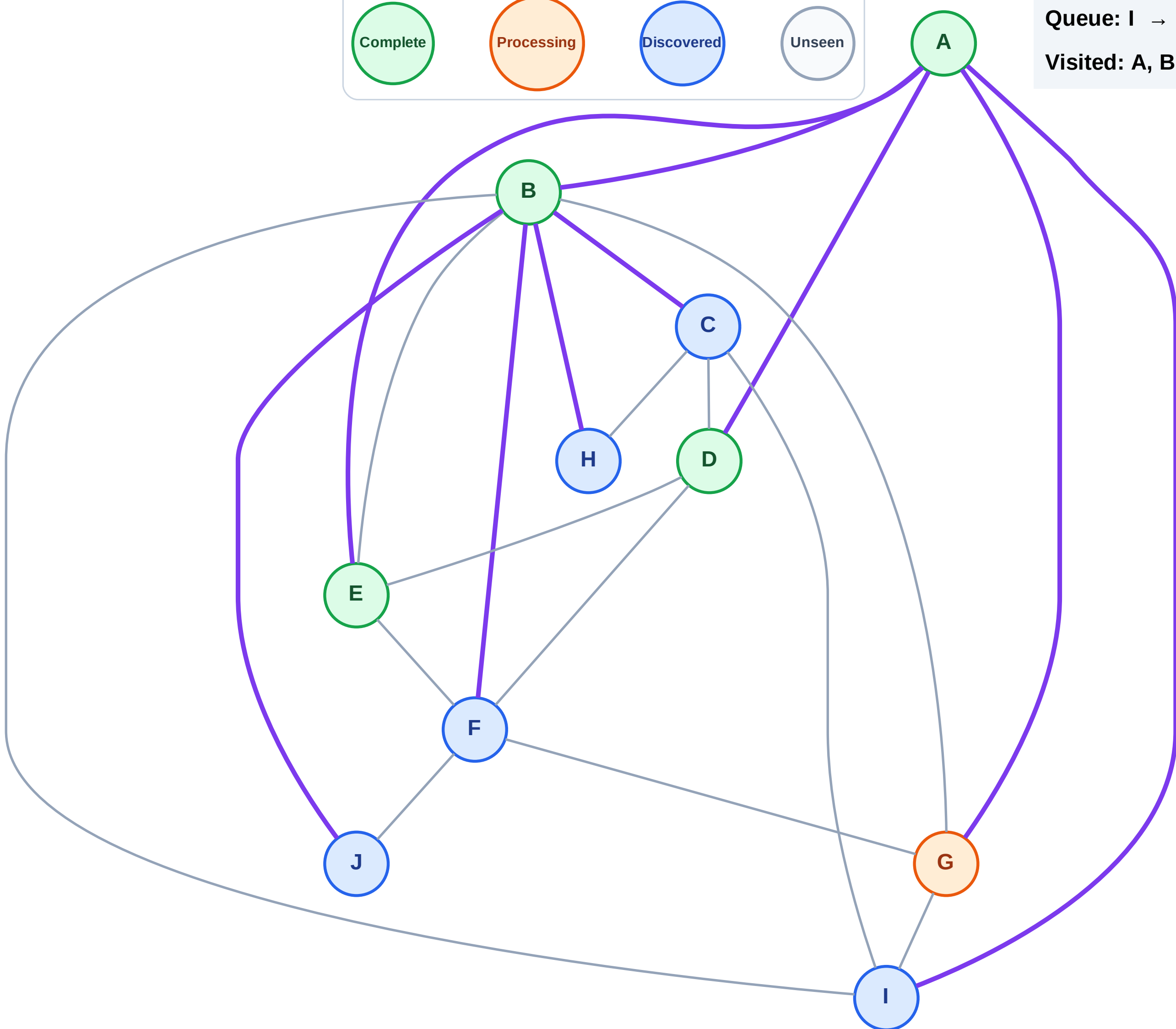
Step 10: Process node G

## Legend



Queue: I → C → F → H → J

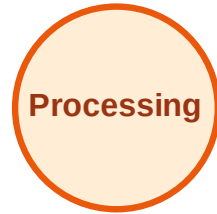
Visited: A, B, D, E



# BFS Traversal

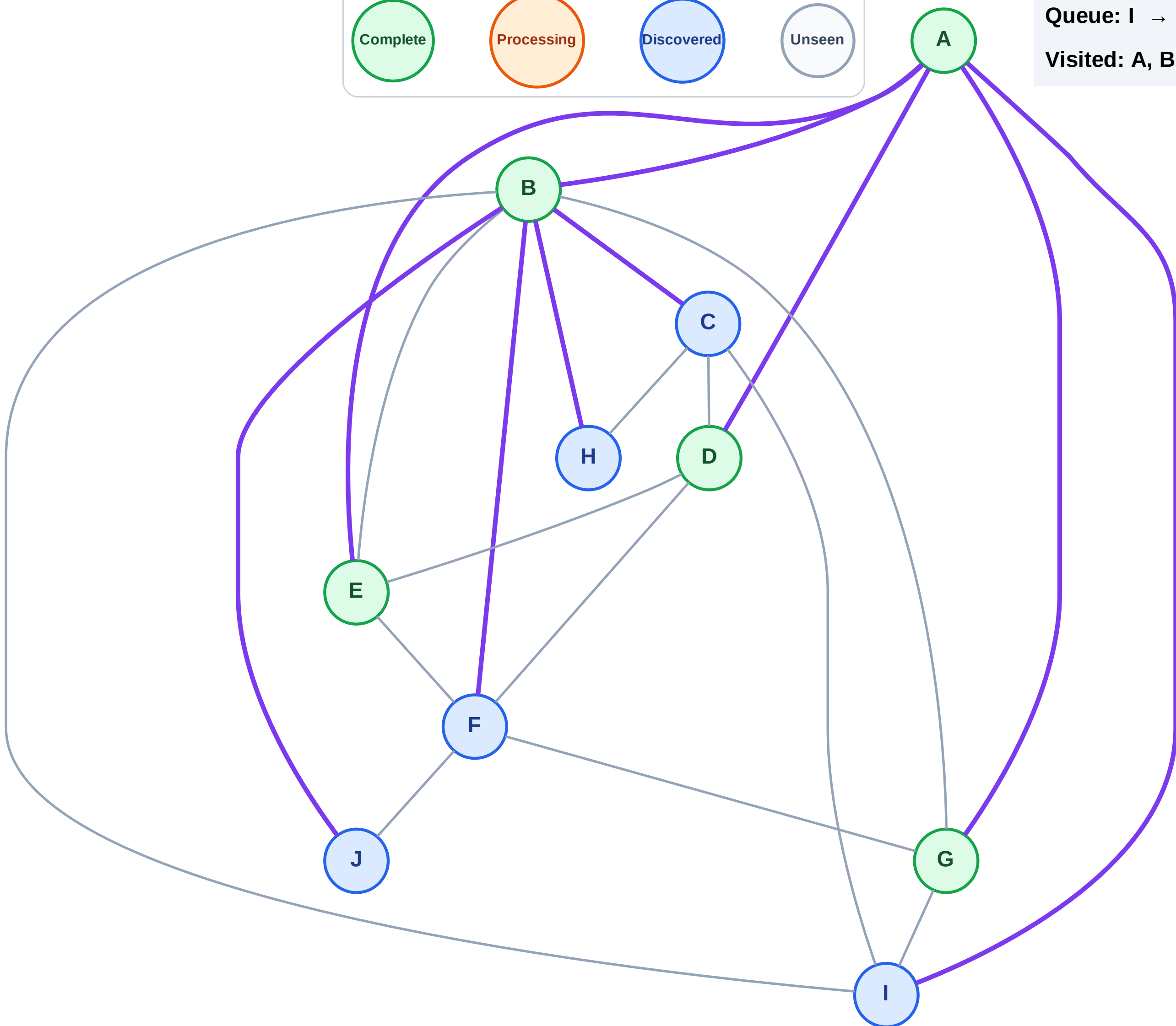
Step 11: No new neighbors from G

## Legend



Queue: I → C → F → H → J

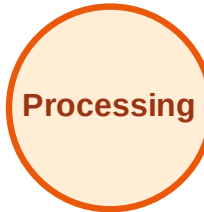
Visited: A, B, D, E, G



# BFS Traversal

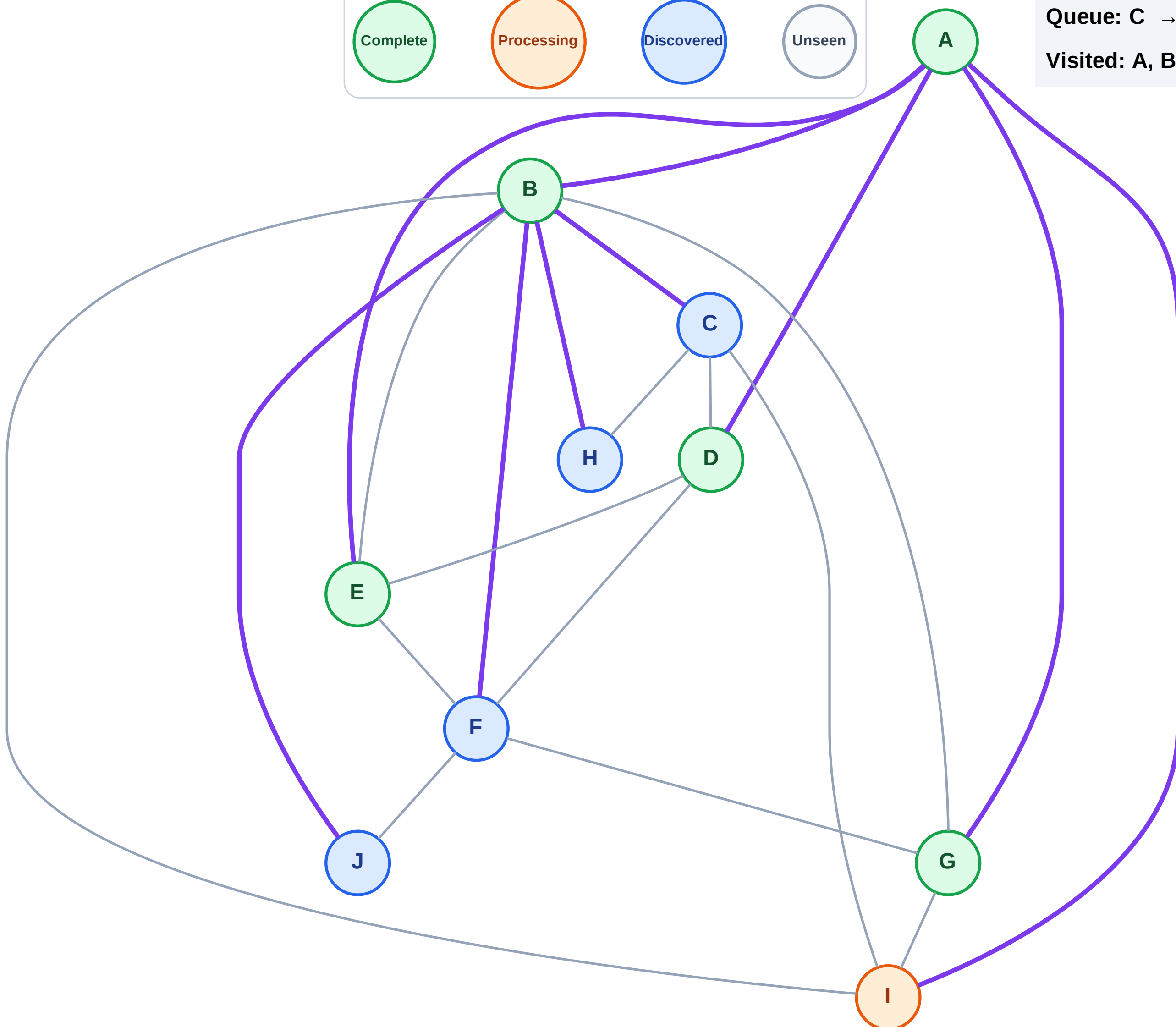
Step 12: Process node I

## Legend



Queue: C → F → H → J

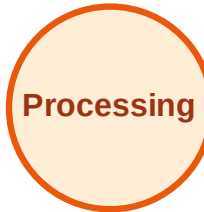
Visited: A, B, D, E, G



# BFS Traversal

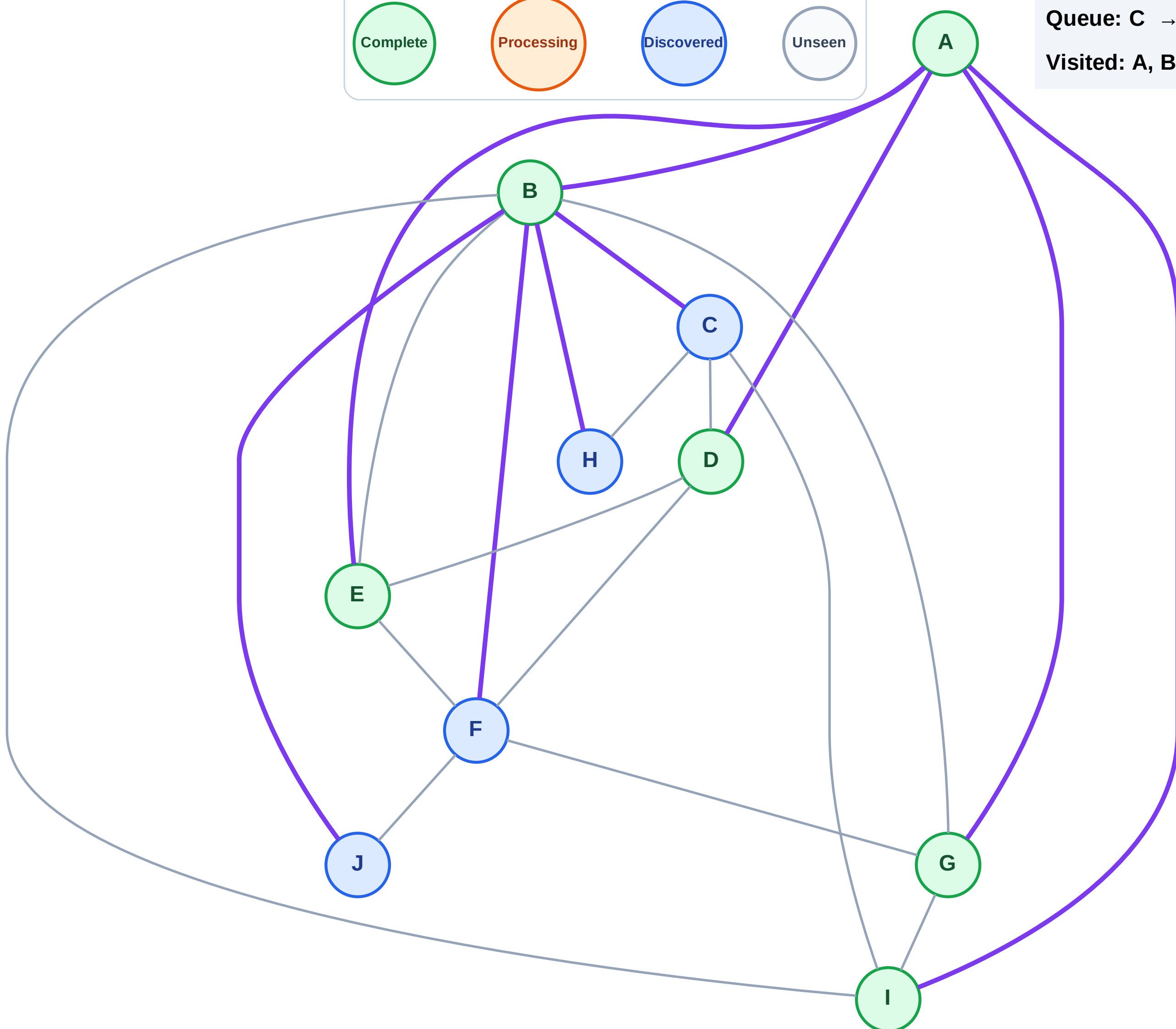
Step 13: No new neighbors from I

## Legend



Queue: C → F → H → J

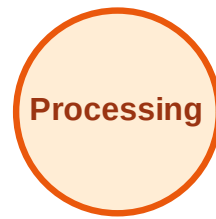
Visited: A, B, D, E, G, I



# BFS Traversal

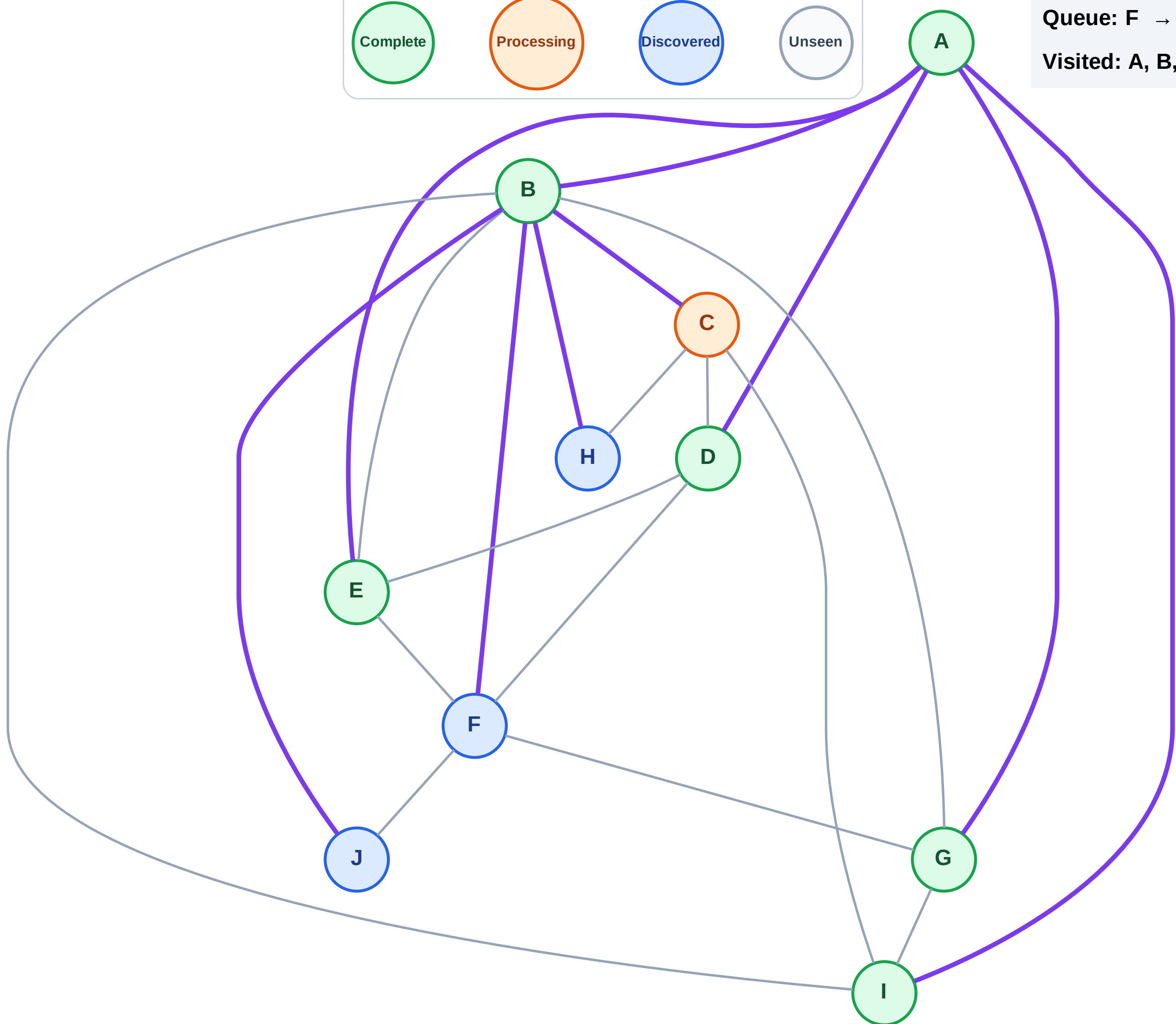
Step 14: Process node C

## Legend



Queue: F → H → J

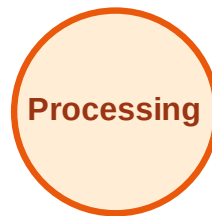
Visited: A, B, D, E, G, I



# BFS Traversal

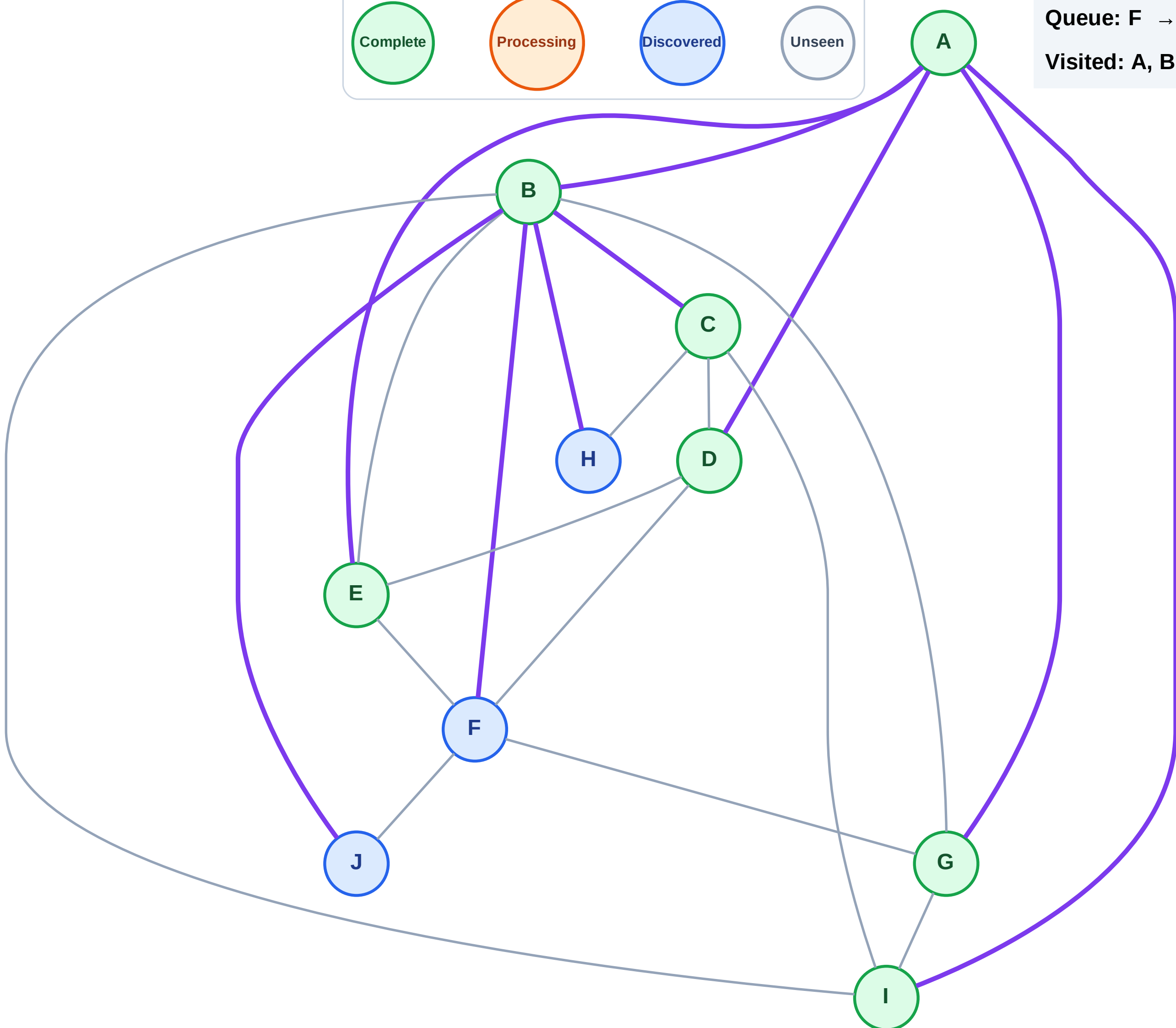
Step 15: No new neighbors from C

## Legend



Queue: F → H → J

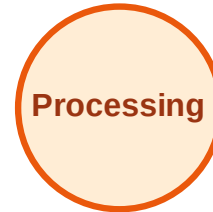
Visited: A, B, C, D, E, G, I



# BFS Traversal

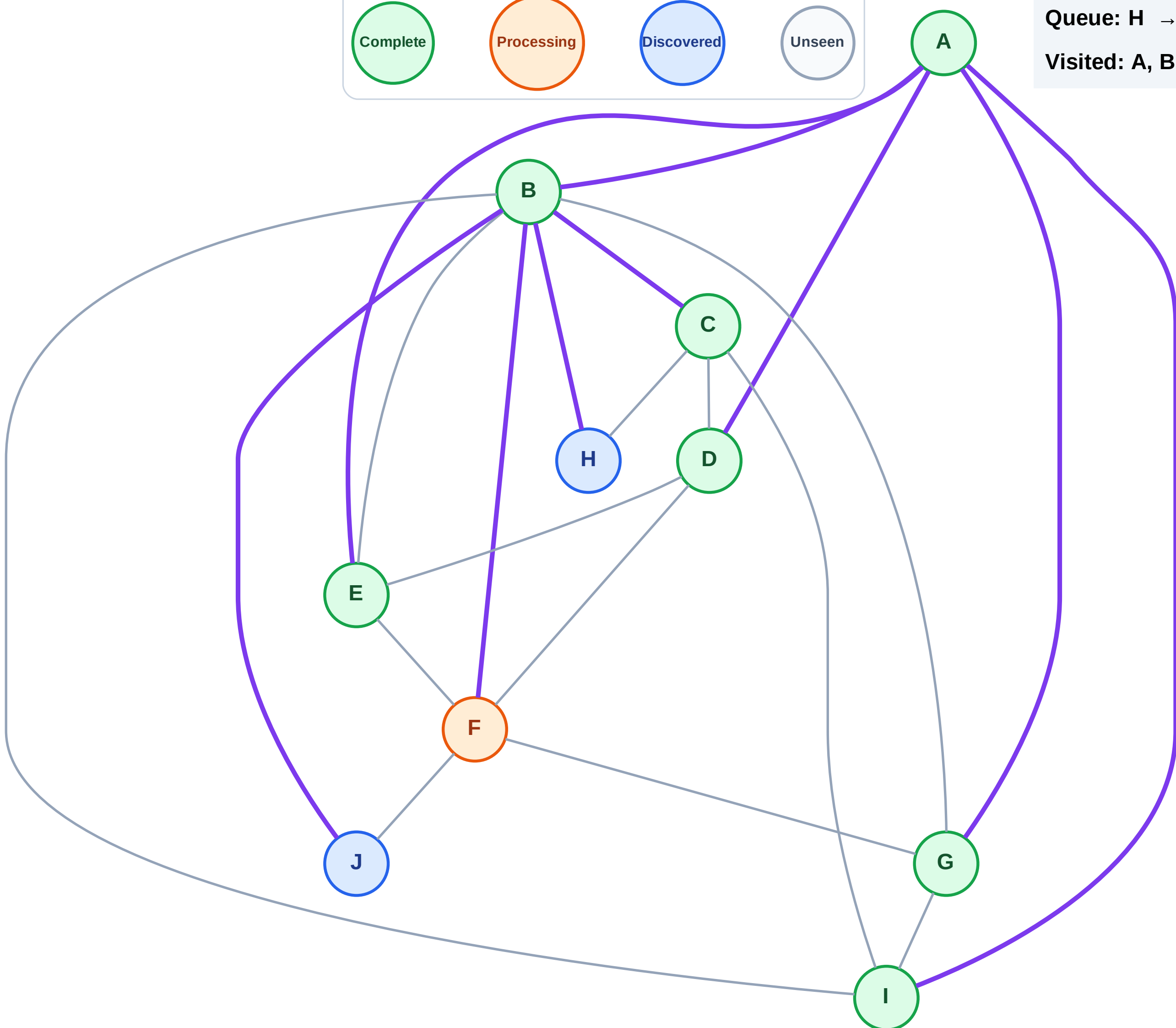
Step 16: Process node F

## Legend



Queue: H → J

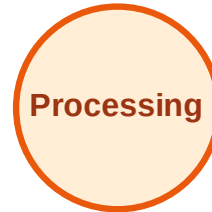
Visited: A, B, C, D, E, G, I



# BFS Traversal

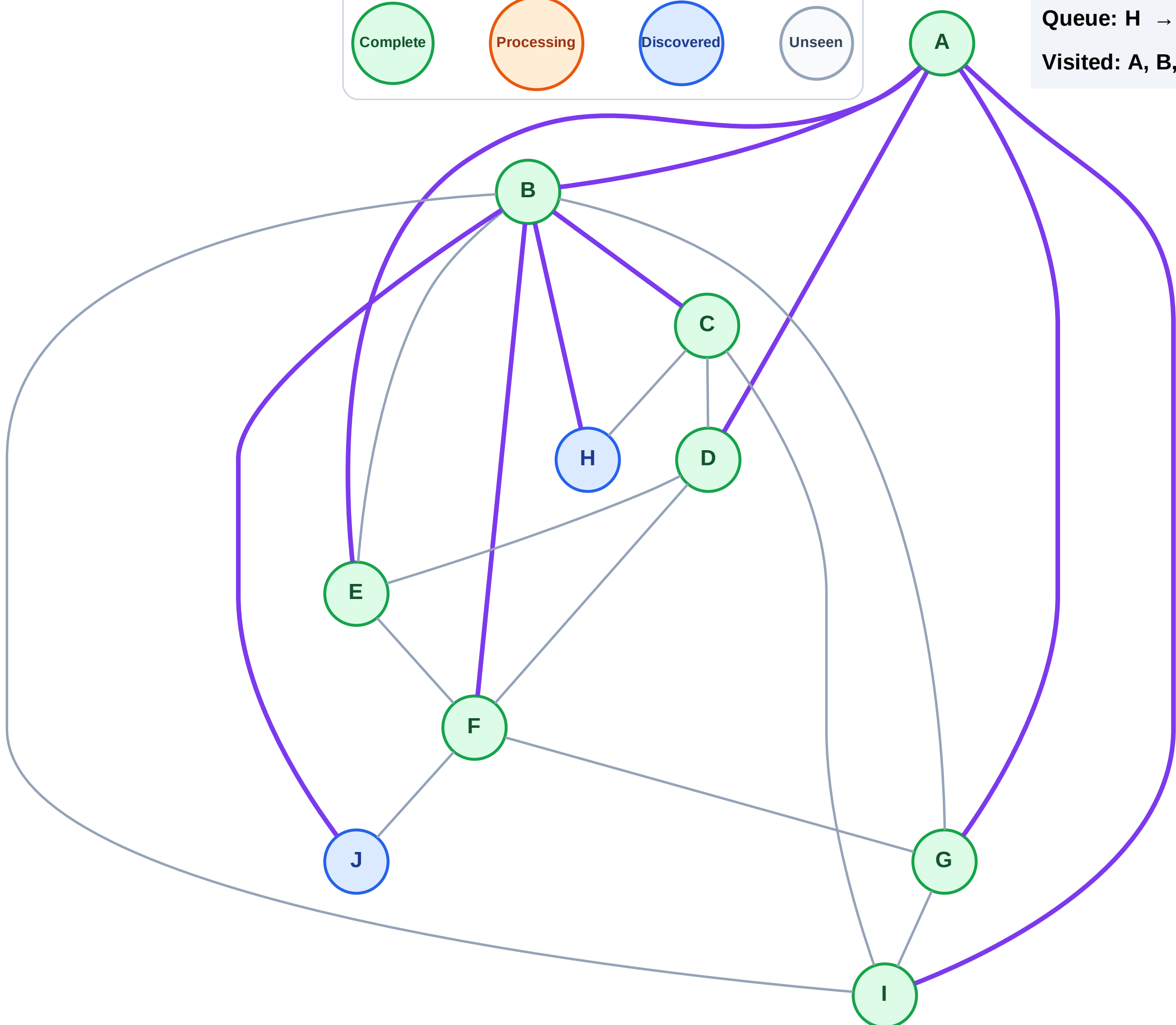
Step 17: No new neighbors from F

## Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



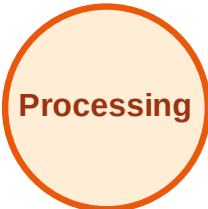
# BFS Traversal

Step 18: Process node H

## Legend



Complete



Processing



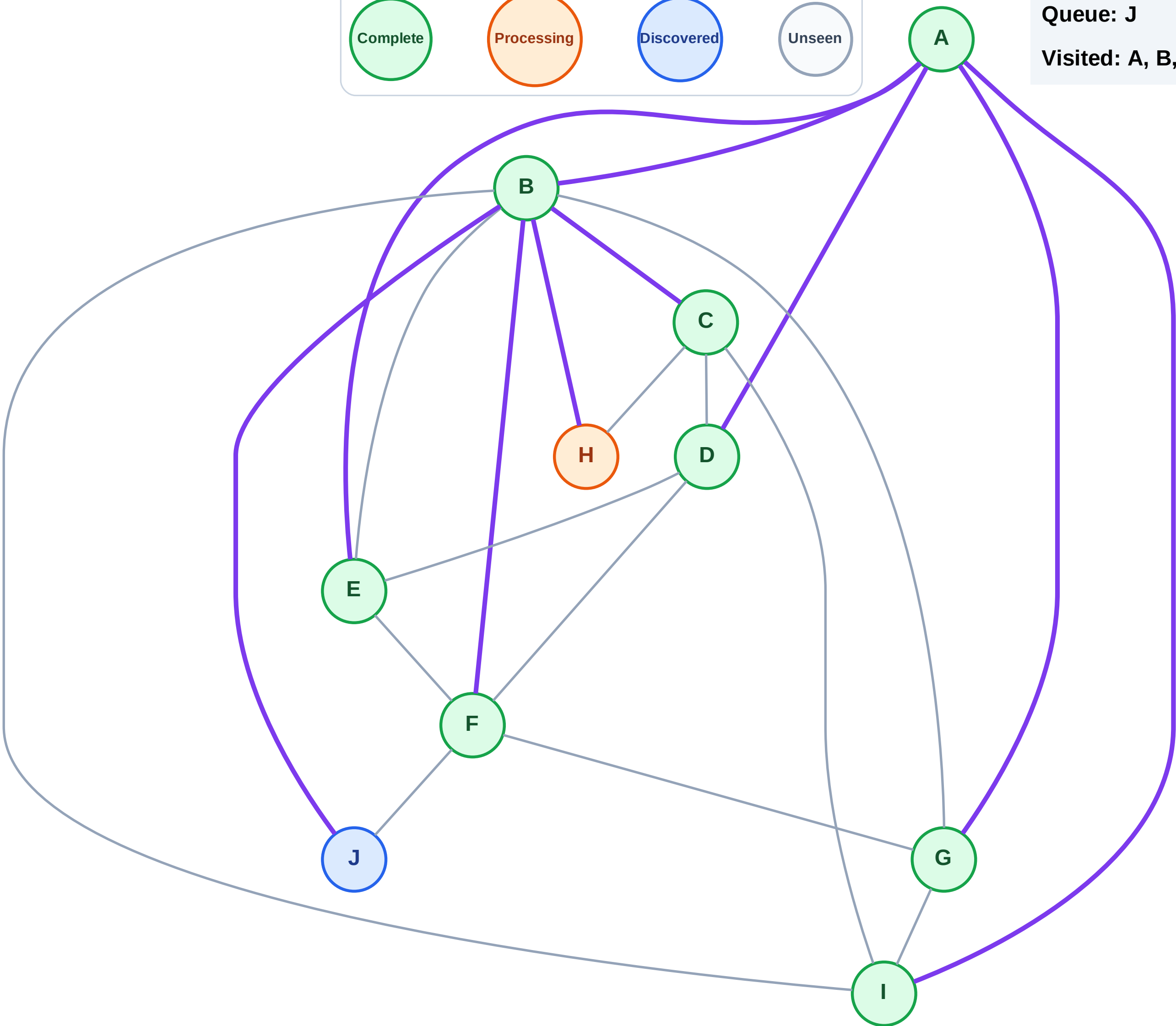
Discovered



Unseen

Queue: J

Visited: A, B, C, D, E, F, G, I



# BFS Traversal

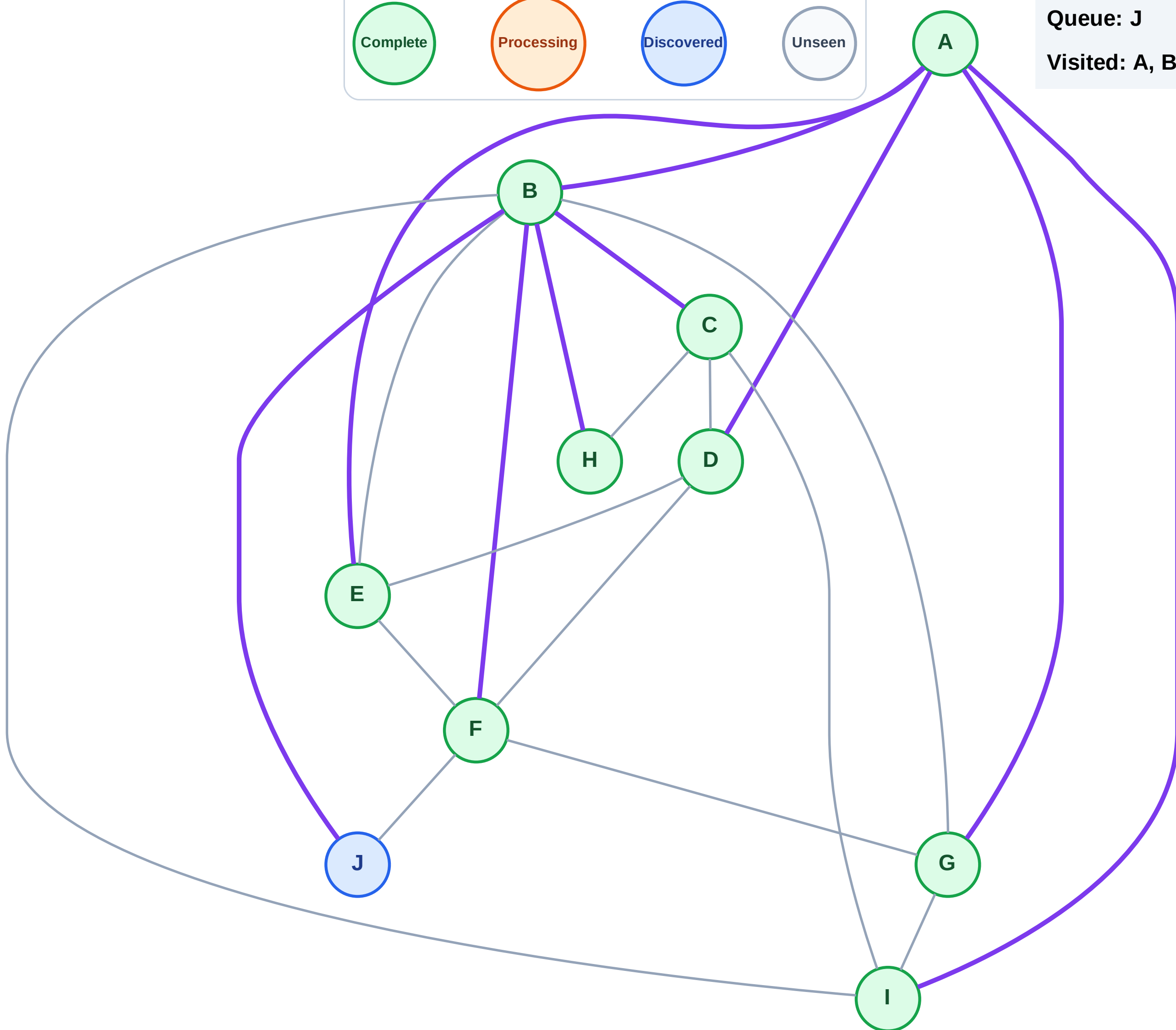
Step 19: No new neighbors from H

## Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

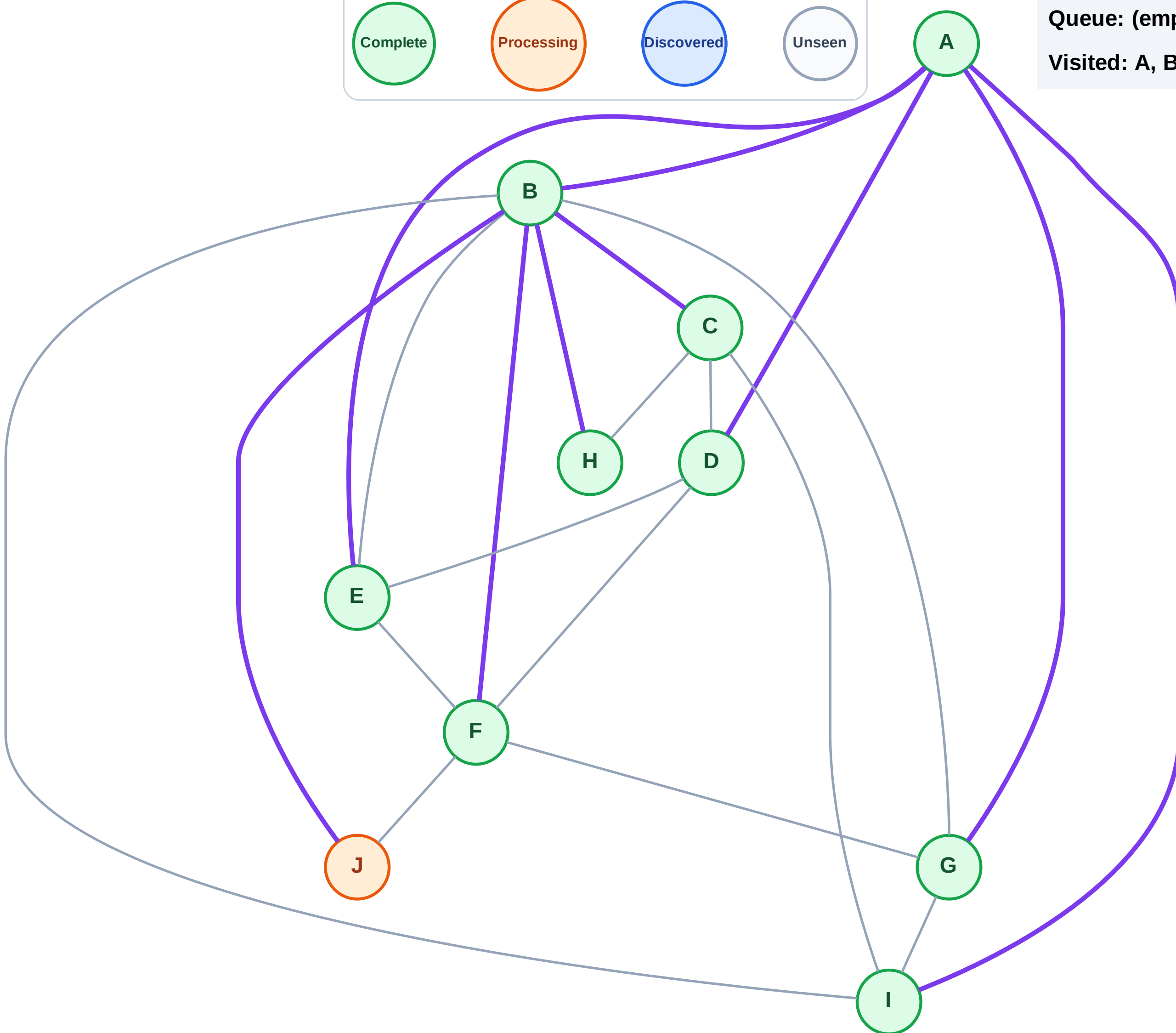
Step 20: Process node J

## Legend



Queue: (empty)

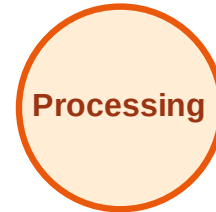
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

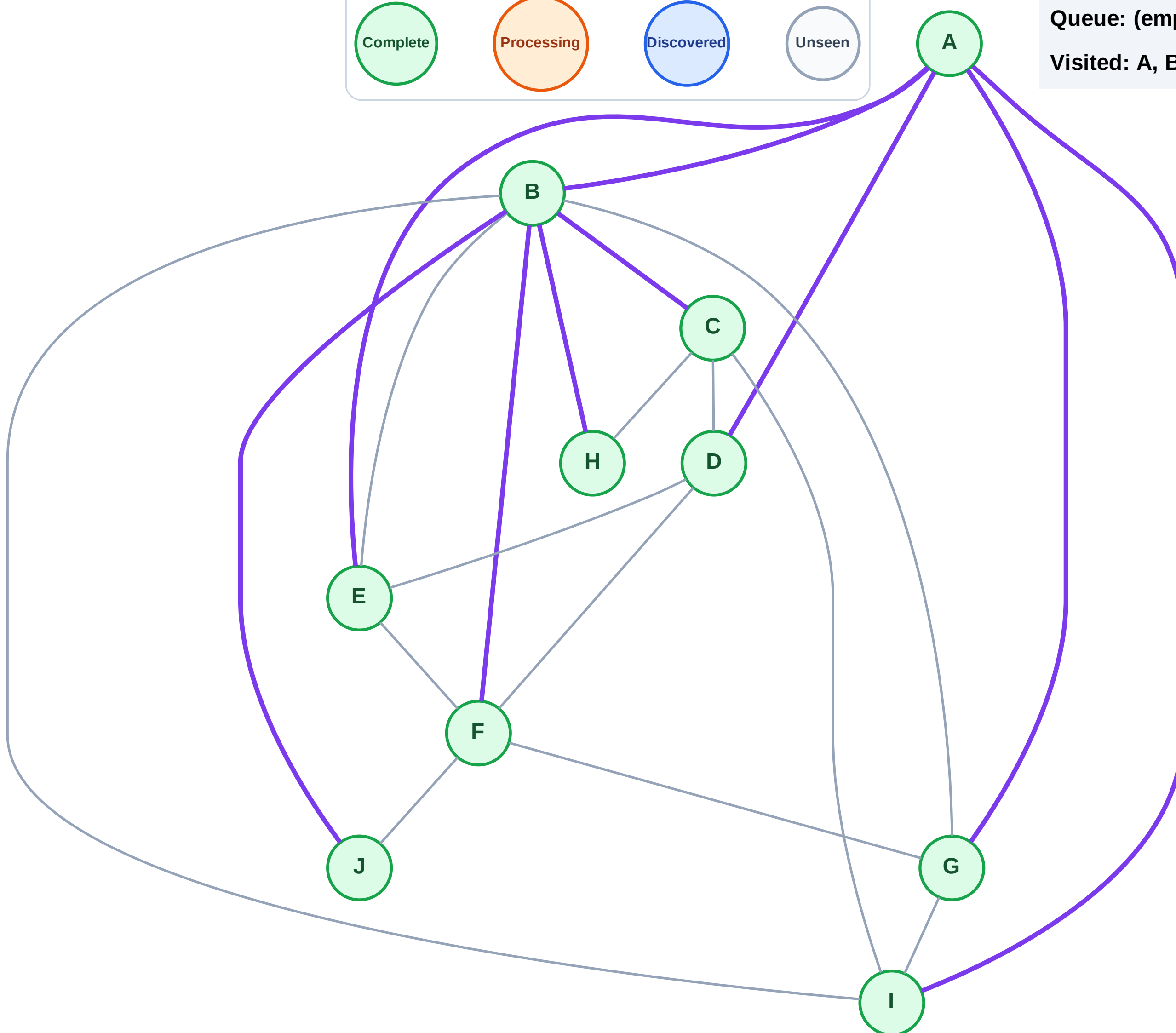
Step 21: No new neighbors from J

## Legend



Queue: (empty)

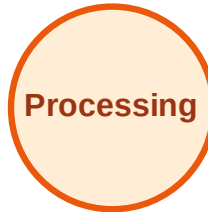
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

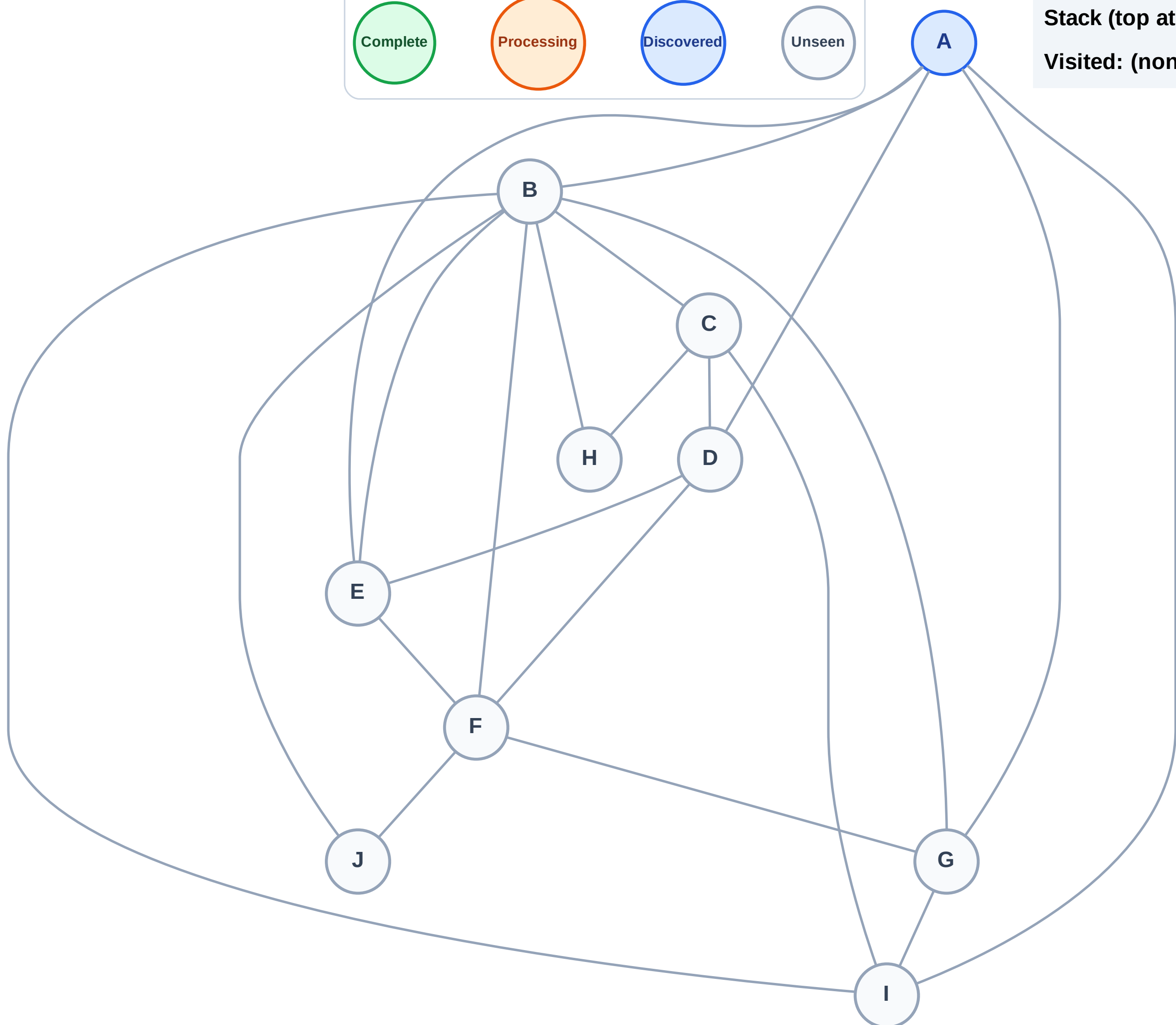
Step 1: Start at A

## Legend



Stack (top at right): A

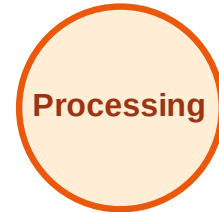
Visited: (none)



# DFS Traversal

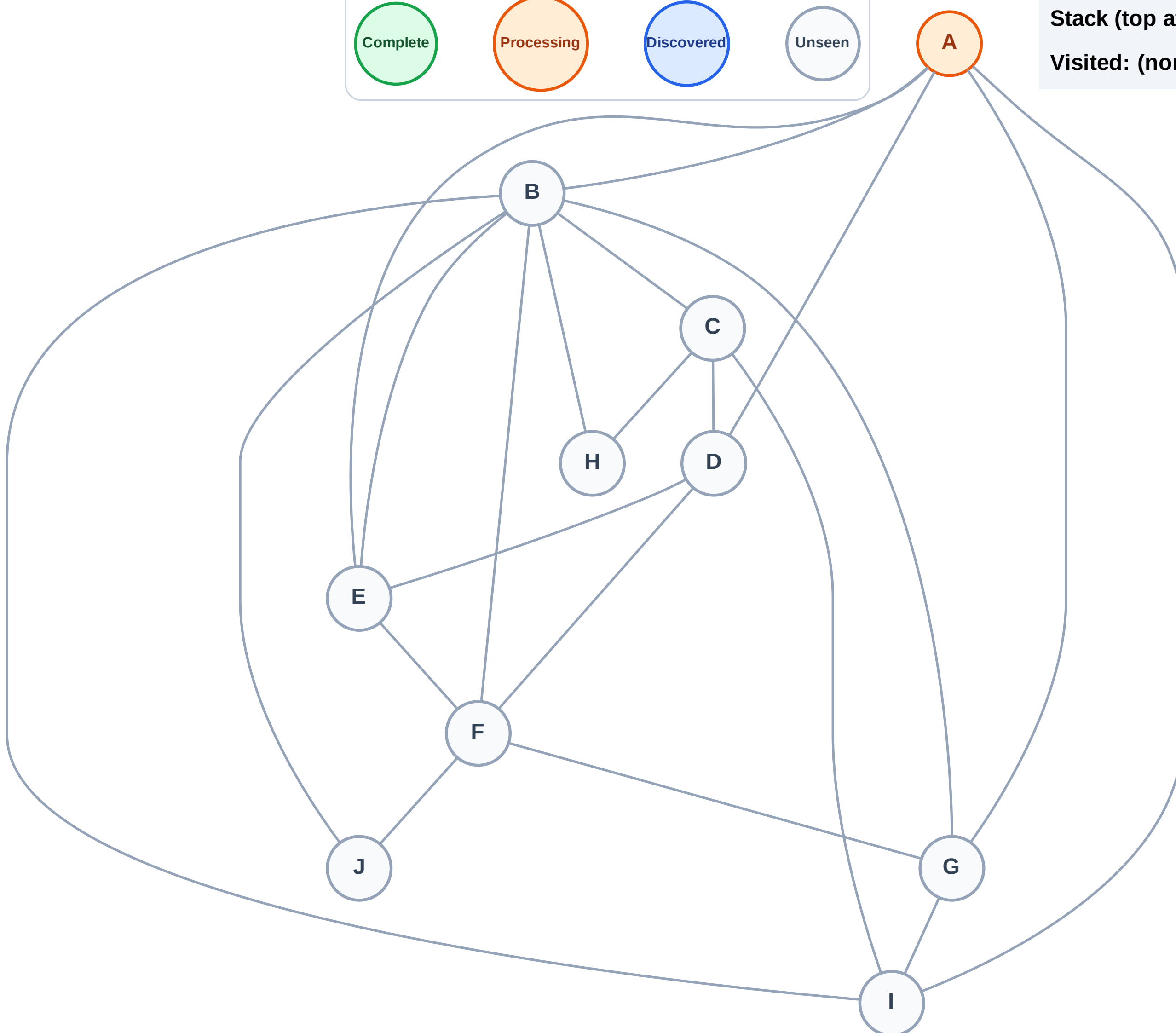
Step 2: Process node A

## Legend



Stack (top at right): (empty)

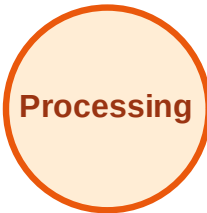
Visited: (none)



# DFS Traversal

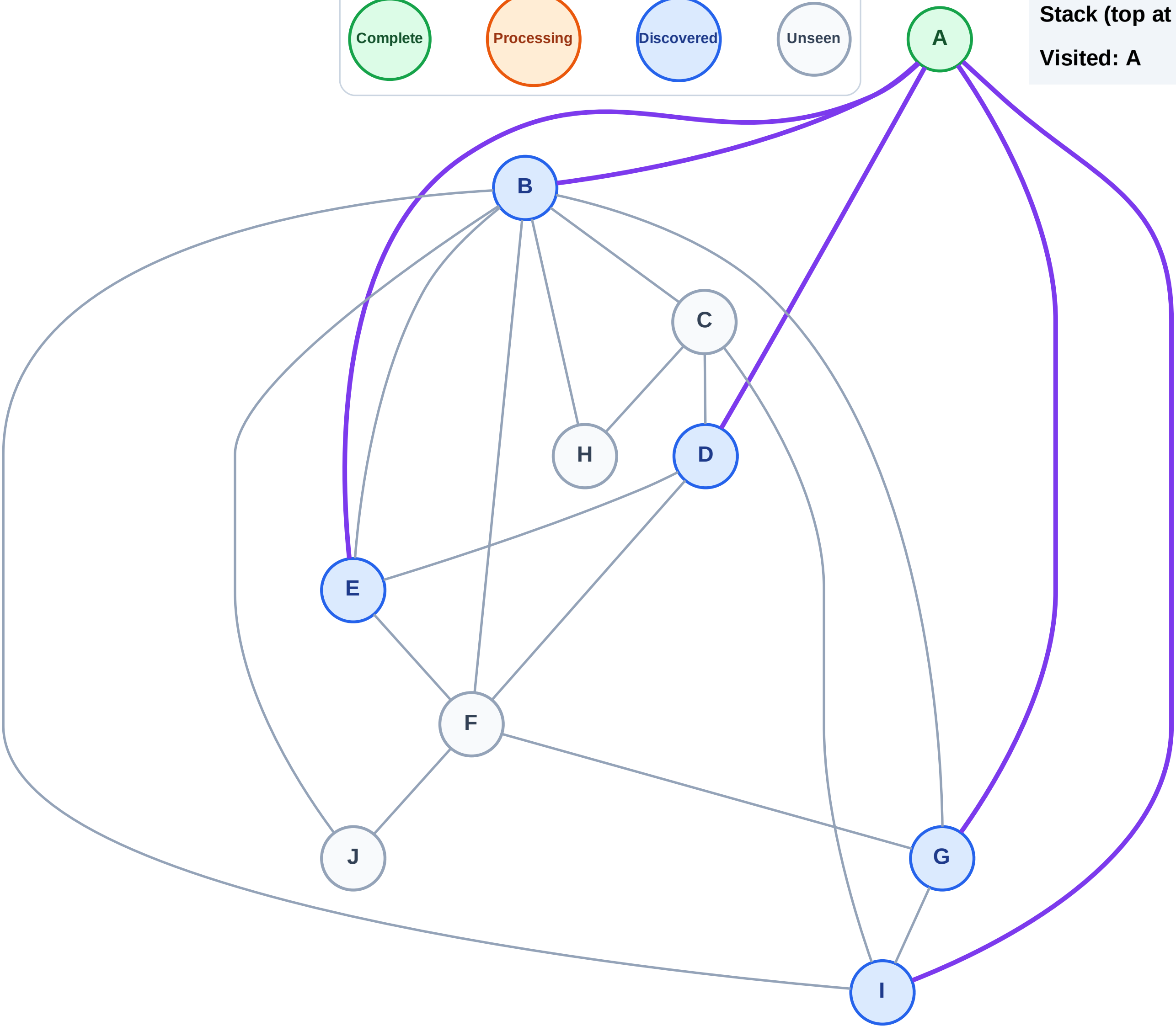
Step 3: Discover I, G, E, D, B from A

## Legend



Stack (top at right): I | G | E | D | B

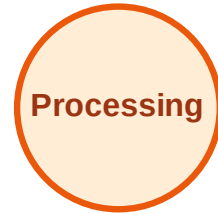
Visited: A



# DFS Traversal

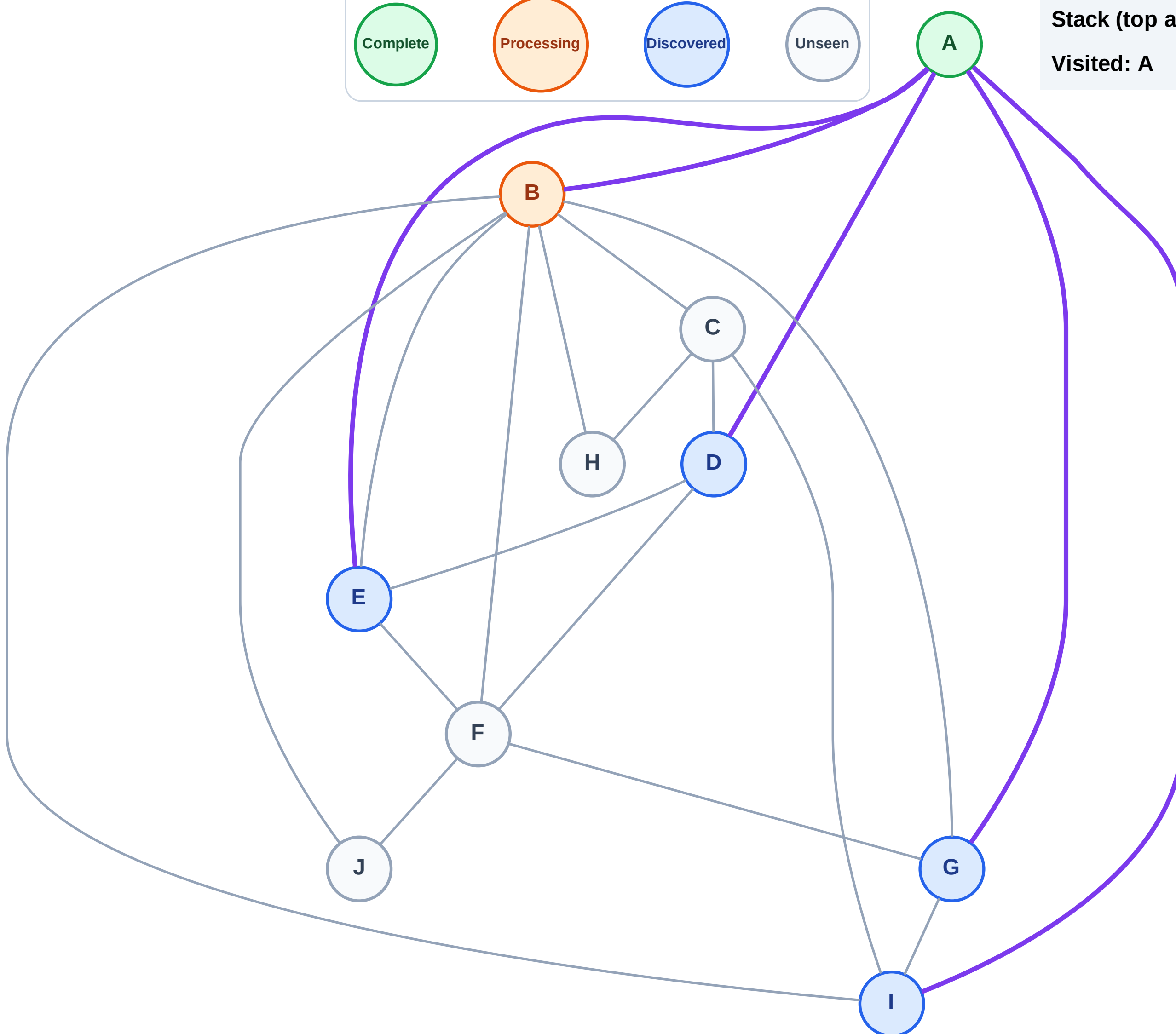
Step 4: Process node B

## Legend



Stack (top at right): I | G | E | D

Visited: A



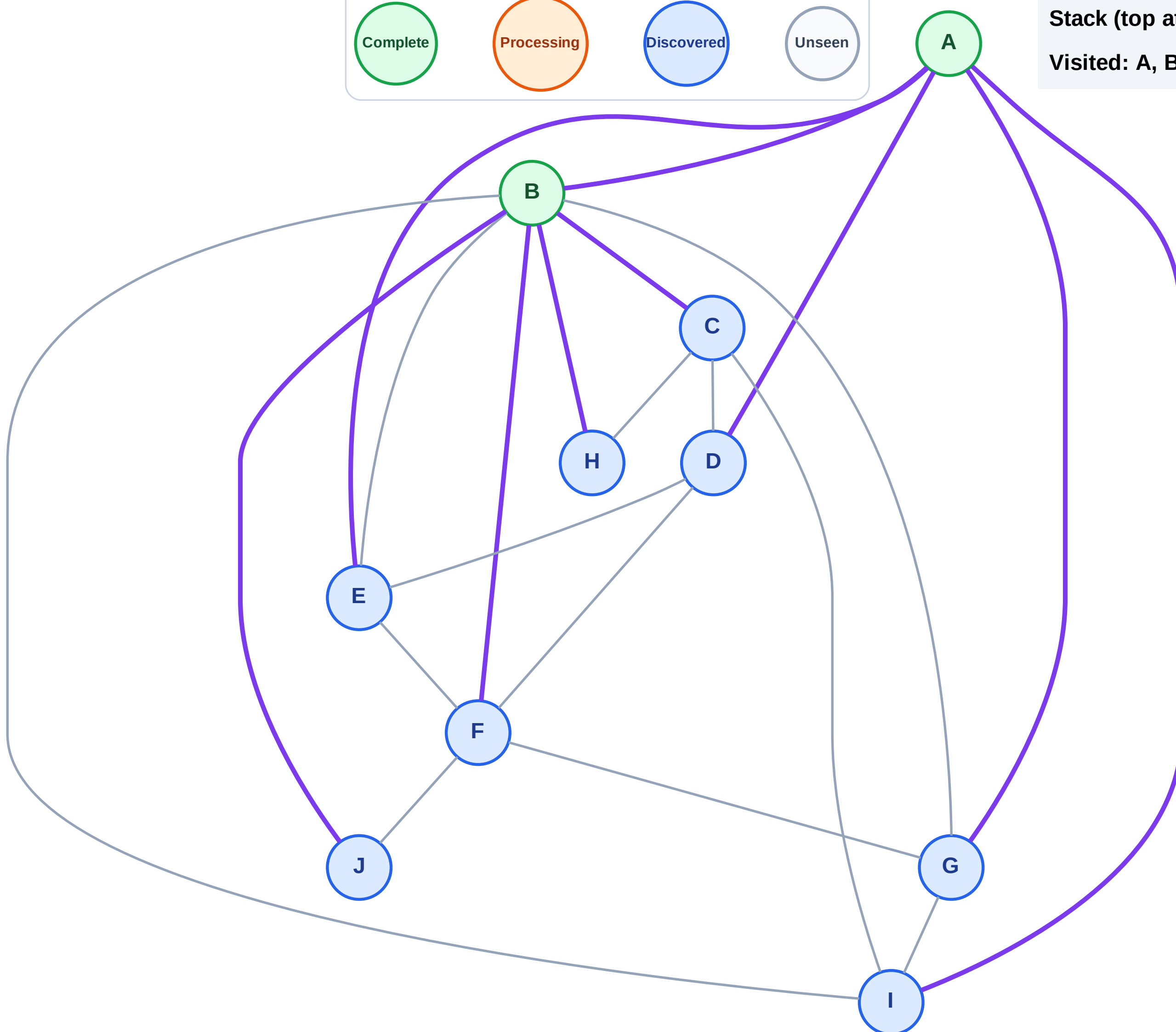
# DFS Traversal

Step 5: Discover J, H, F, C from B

## Legend



Stack (top at right): I | G | E | D | J | H | F | C  
Visited: A, B



# DFS Traversal

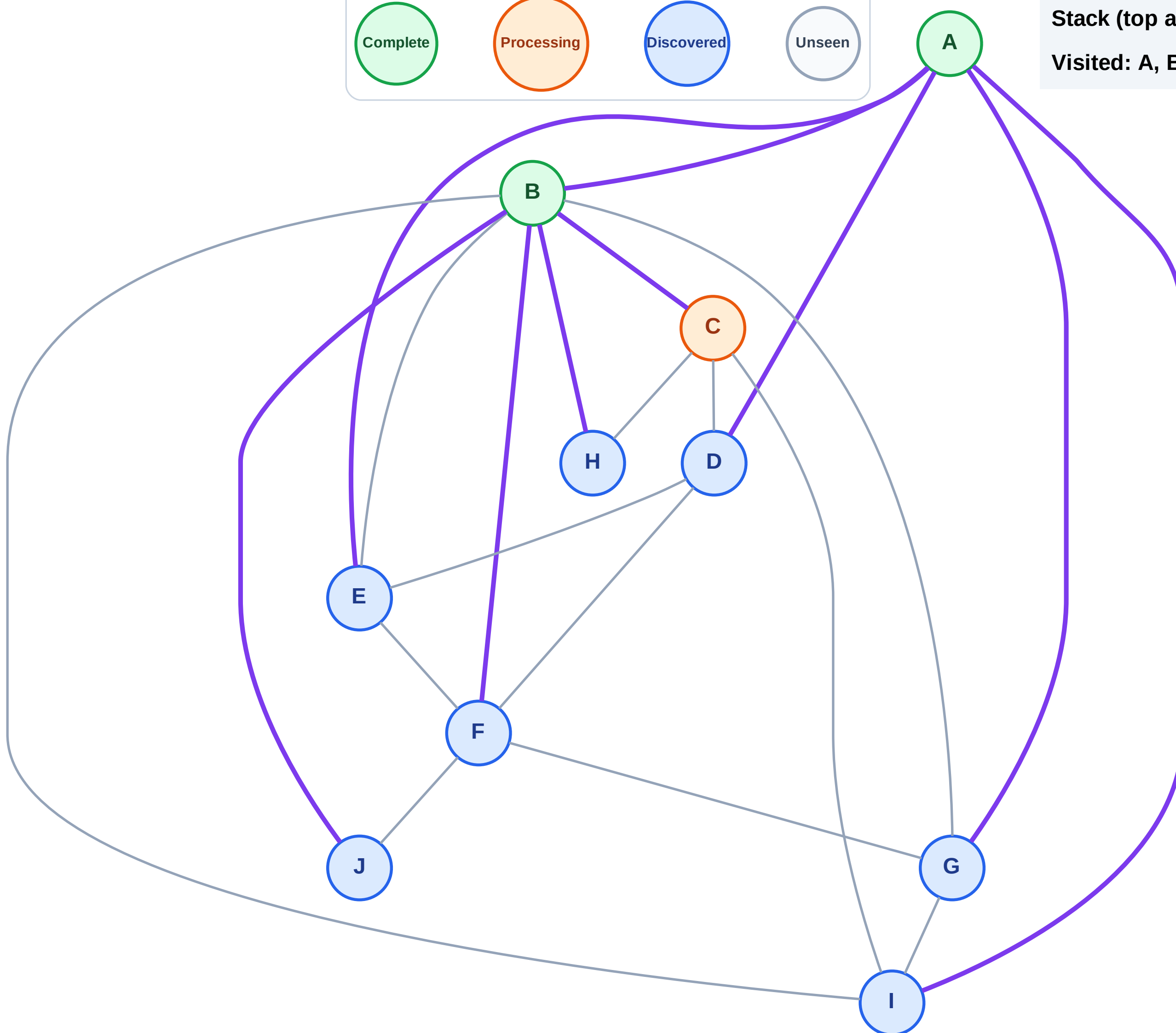
Step 6: Process node C

## Legend



Stack (top at right): I | G | E | D | J | H | F

Visited: A, B



## Step 7: No new neighbors from C

## Step 7: No new neighbors from C

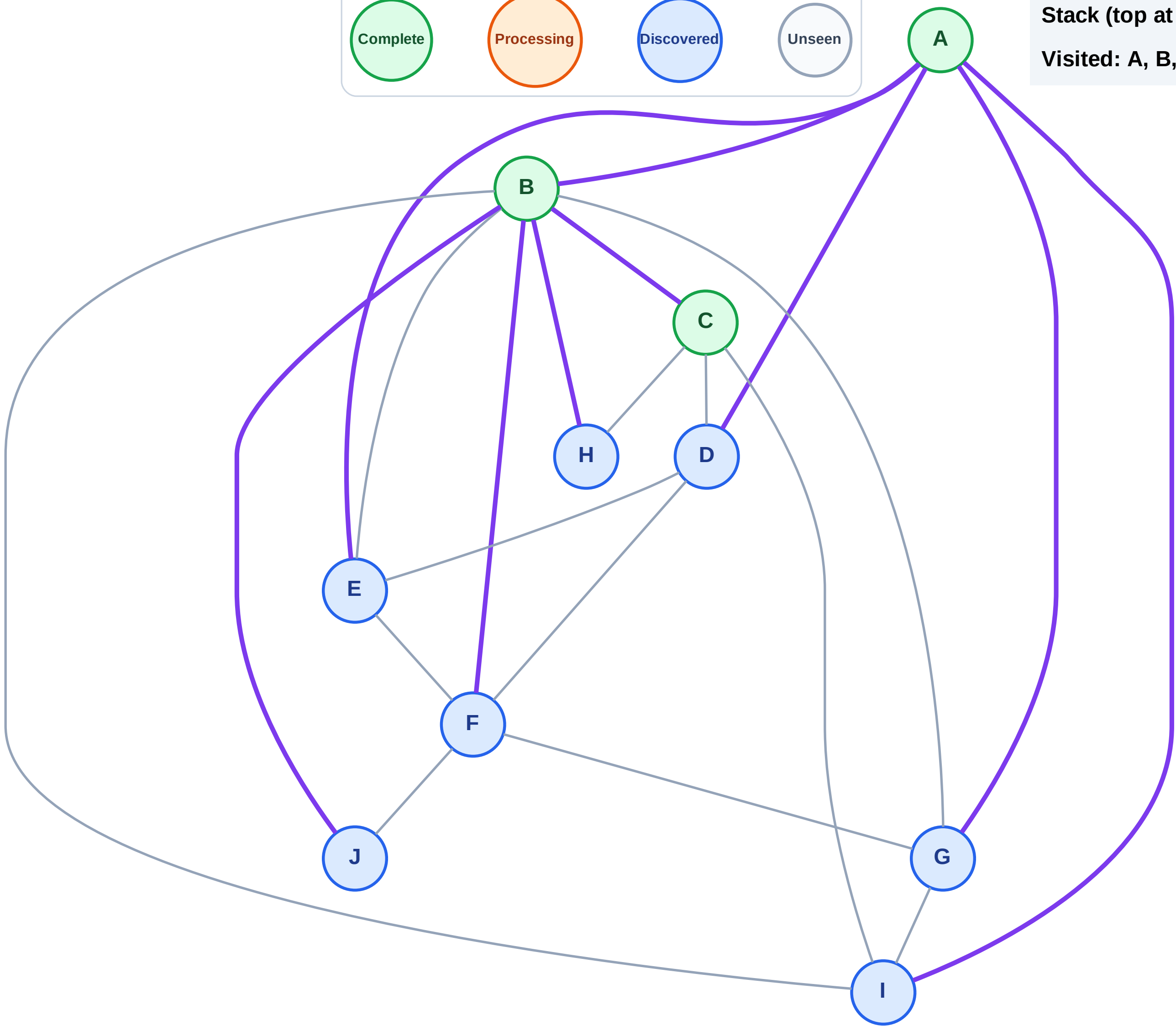
```

graph LR
    Complete((Complete)) --> Processing((Processing))
    Processing --> Discovered((Discovered))
    Discovered --> Unseen((Unseen))
  
```

## Processing

## Unseen

**Visited: A, B, C**



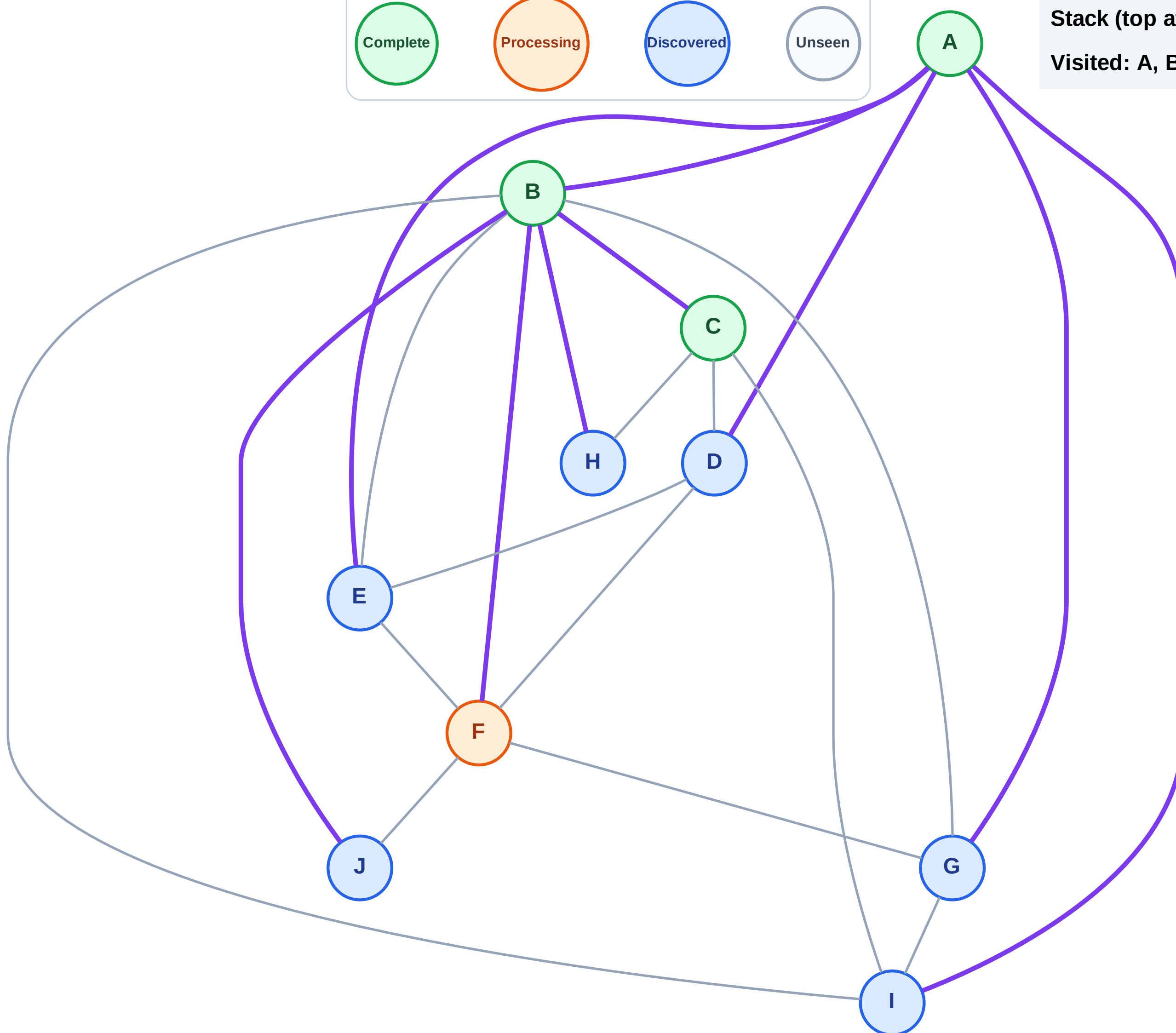
### Step 8: Process node F

---

Complete Processing Discovered Unseen

Unseen

**Visited: A, B, C**



# DFS Traversal

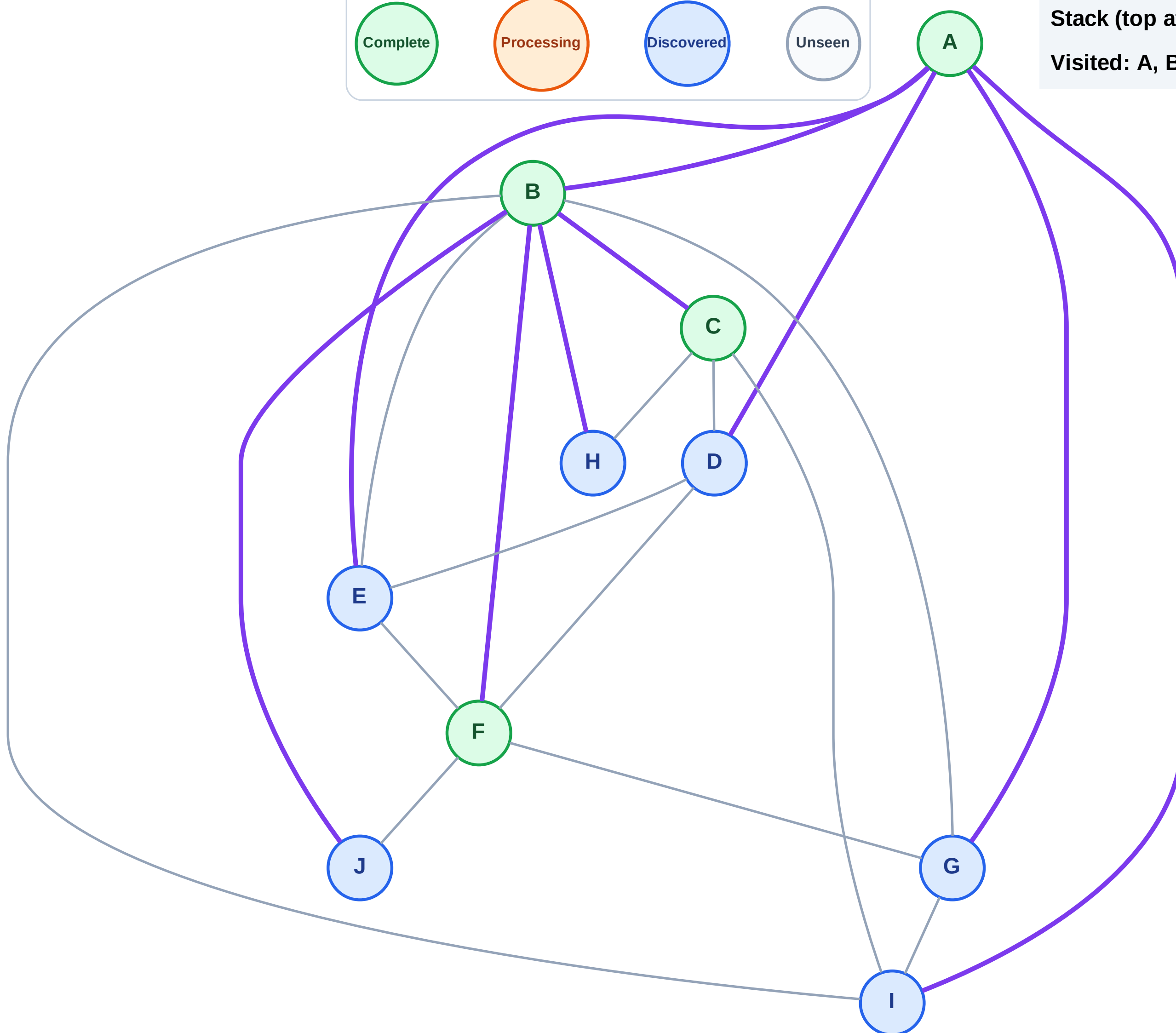
Step 9: No new neighbors from F

## Legend



Stack (top at right): I | G | E | D | J | H

Visited: A, B, C, F



# DFS Traversal

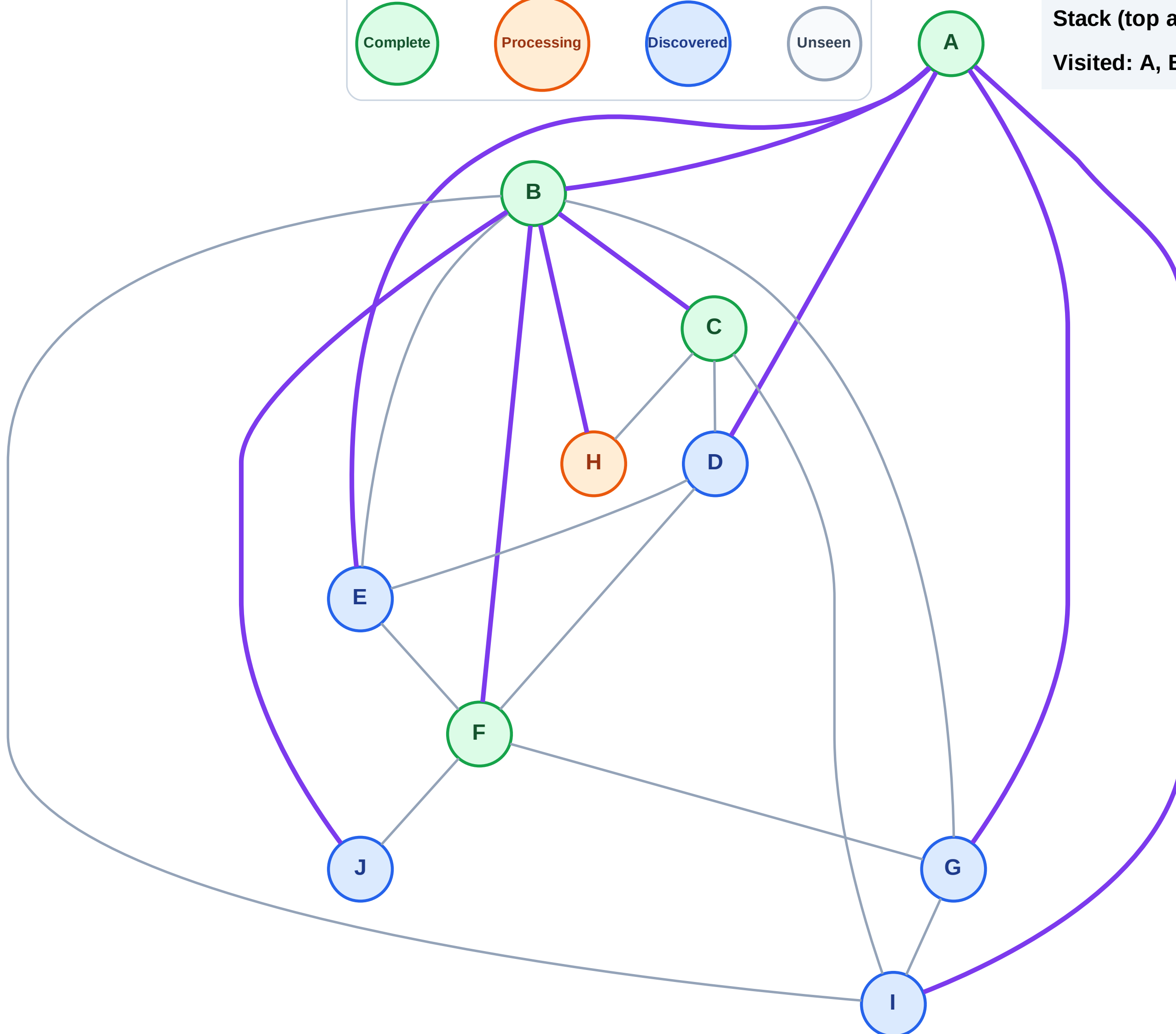
Step 10: Process node H

## Legend



Stack (top at right): I | G | E | D | J

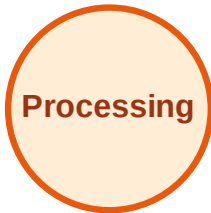
Visited: A, B, C, F



# DFS Traversal

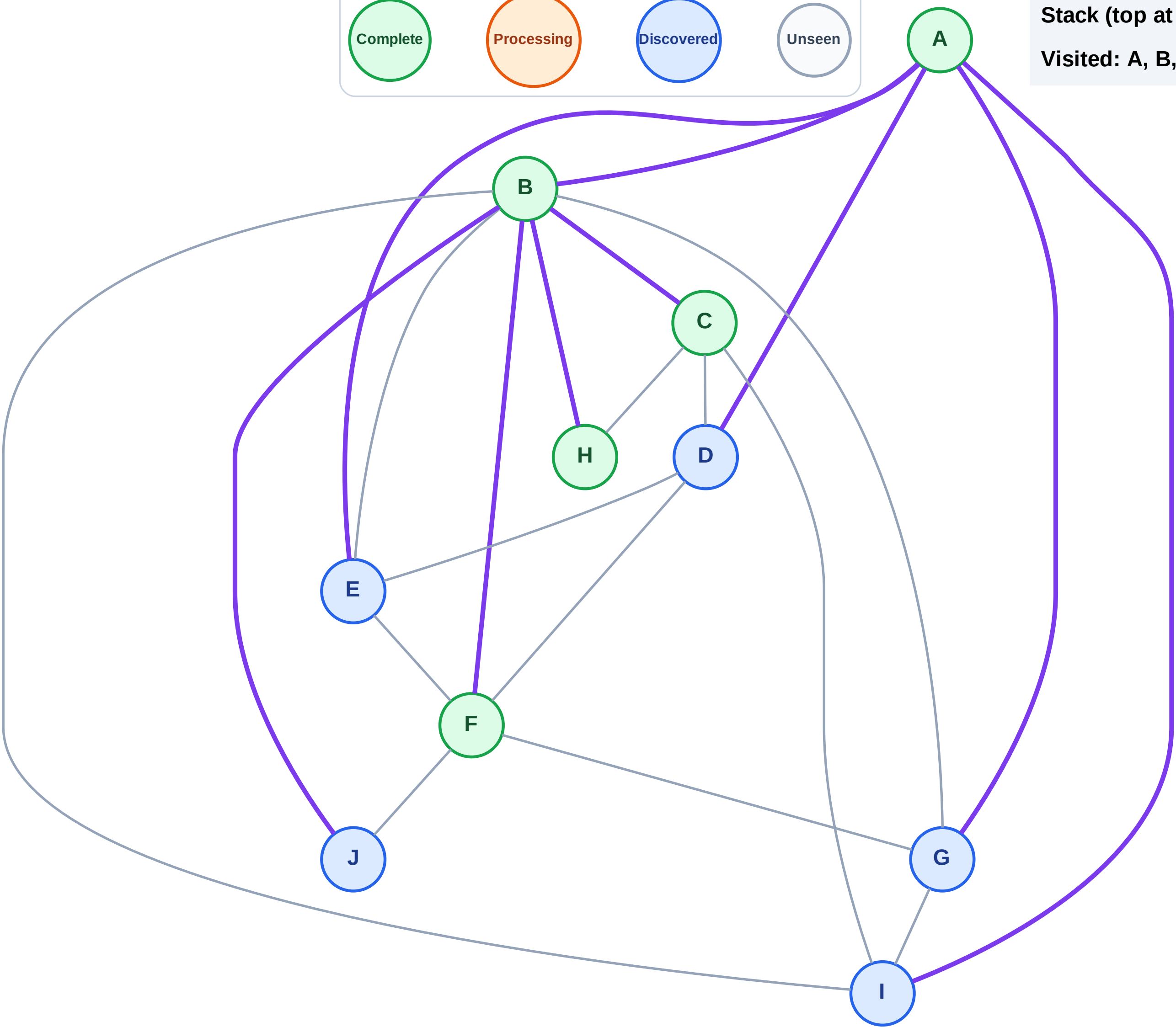
Step 11: No new neighbors from H

## Legend



Stack (top at right): I | G | E | D | J

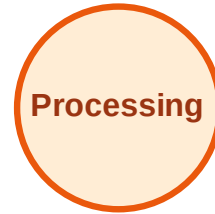
Visited: A, B, C, F, H



# DFS Traversal

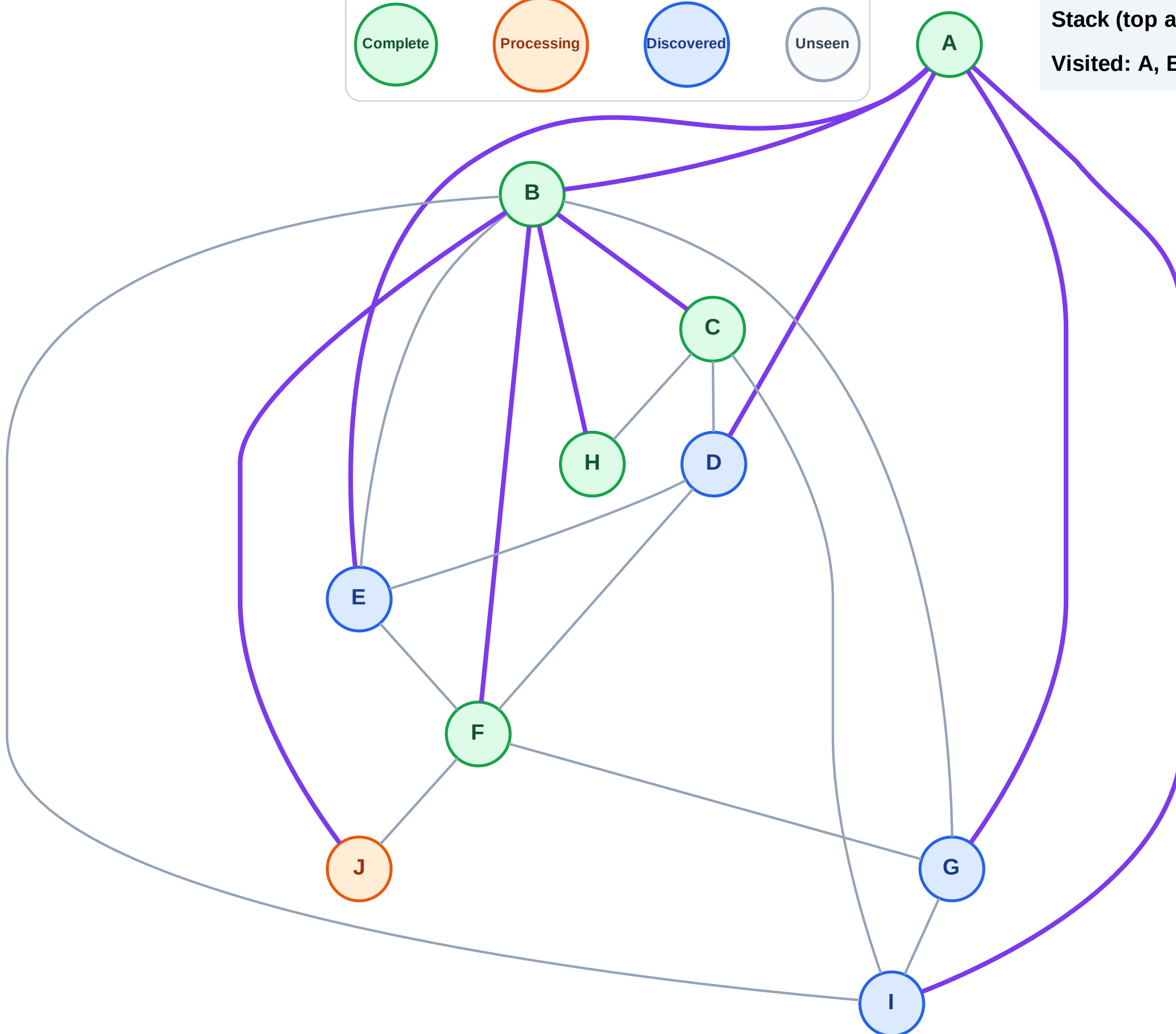
Step 12: Process node J

## Legend



Stack (top at right): I | G | E | D

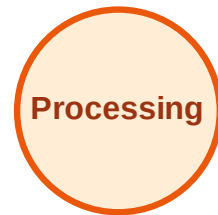
Visited: A, B, C, F, H



# DFS Traversal

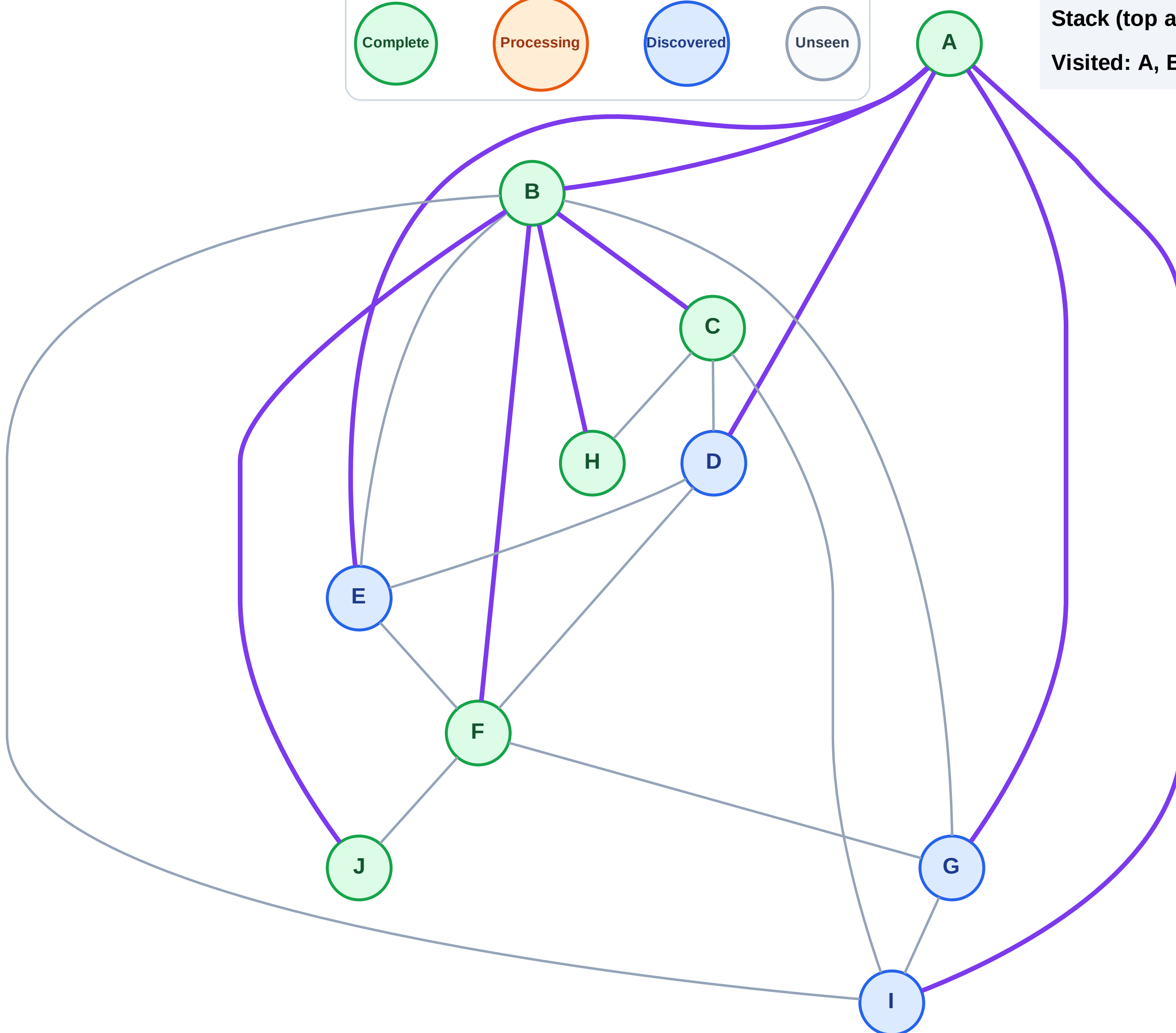
Step 13: No new neighbors from J

## Legend



Stack (top at right): I | G | E | D

Visited: A, B, C, F, H, J



# DFS Traversal

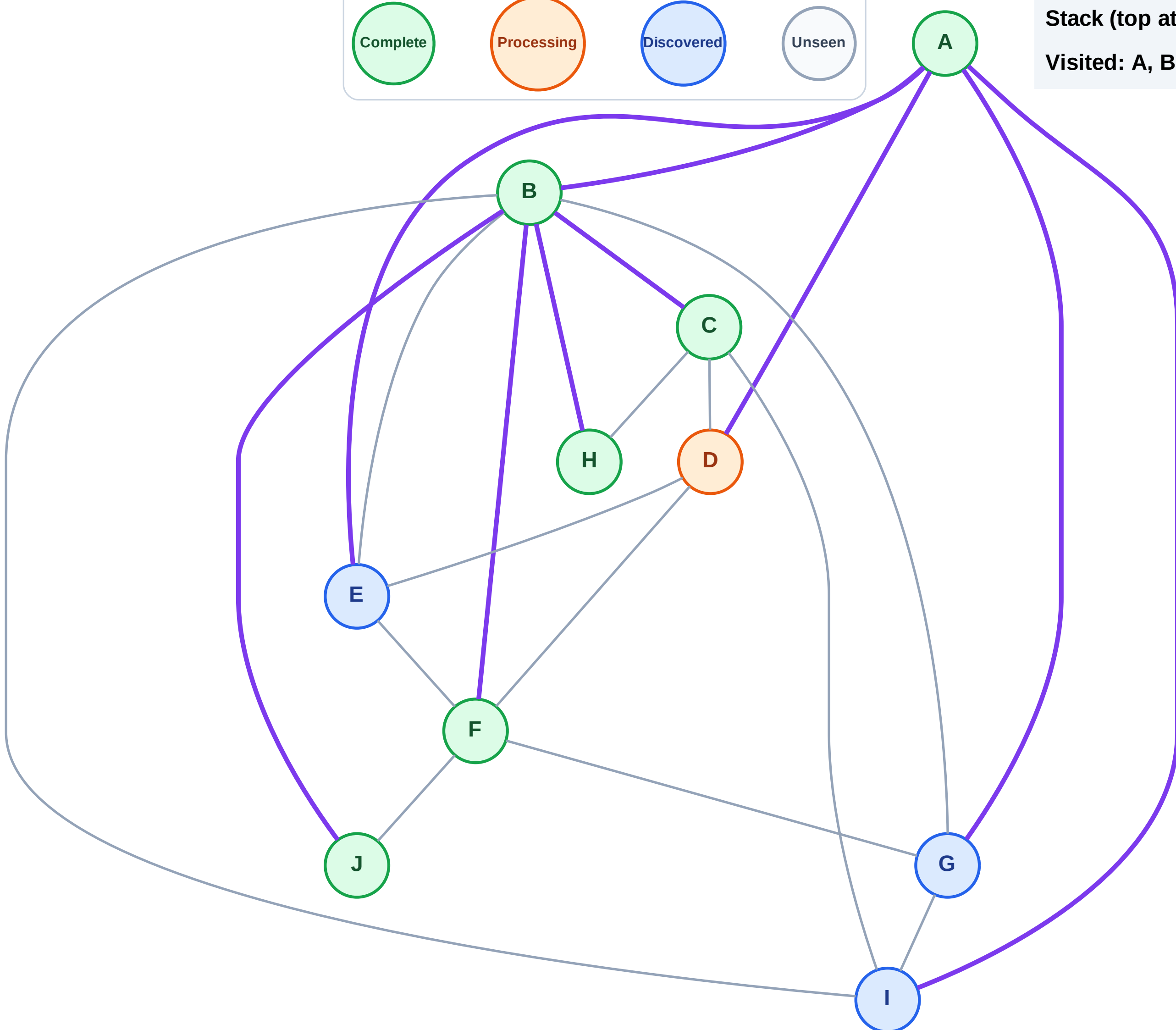
Step 14: Process node D

## Legend



Stack (top at right): I | G | E

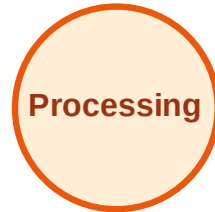
Visited: A, B, C, F, H, J



# DFS Traversal

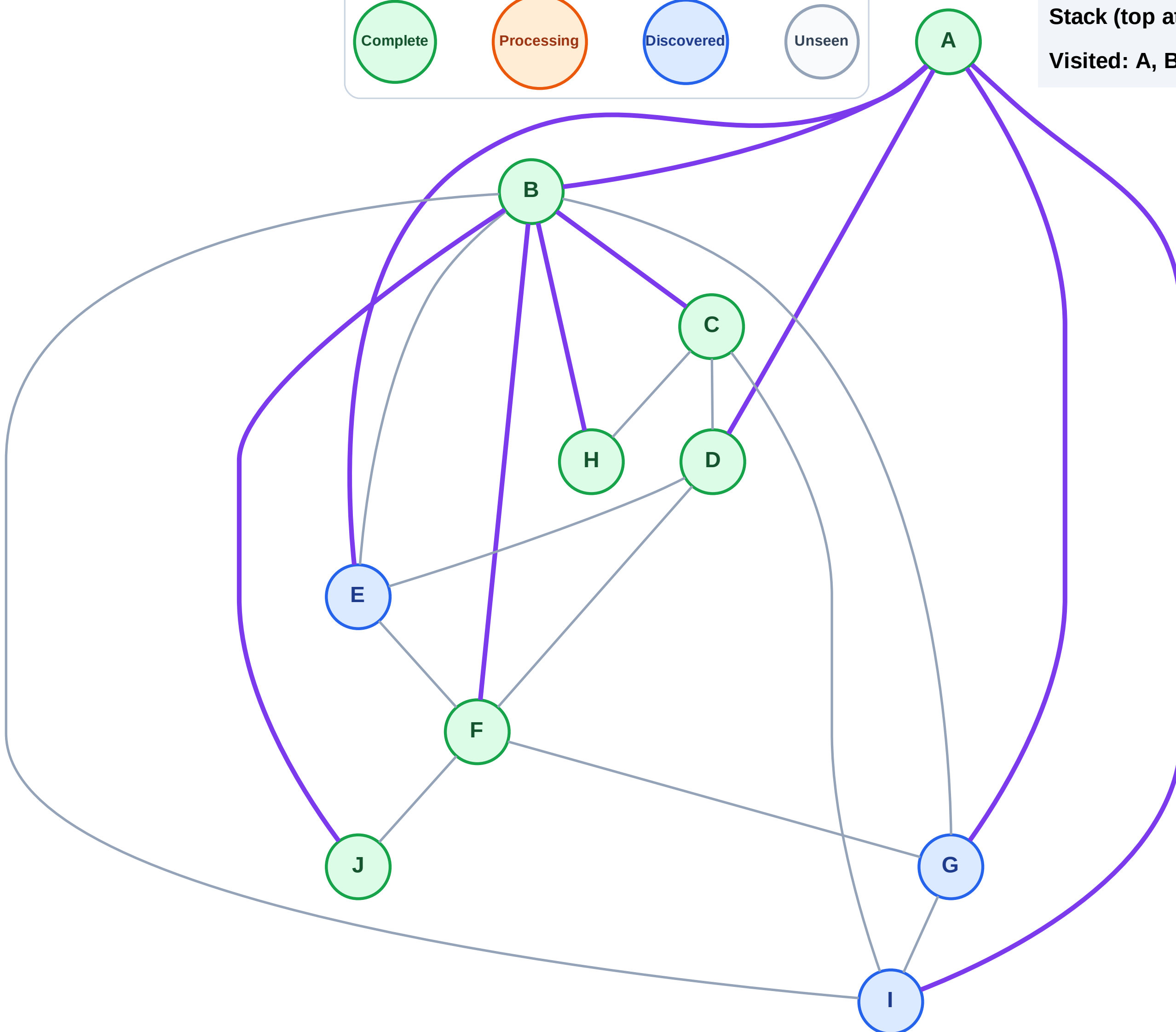
Step 15: No new neighbors from D

## Legend



Stack (top at right): I | G | E

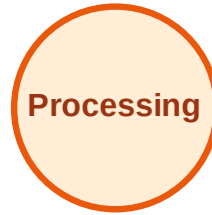
Visited: A, B, C, D, F, H, J



# DFS Traversal

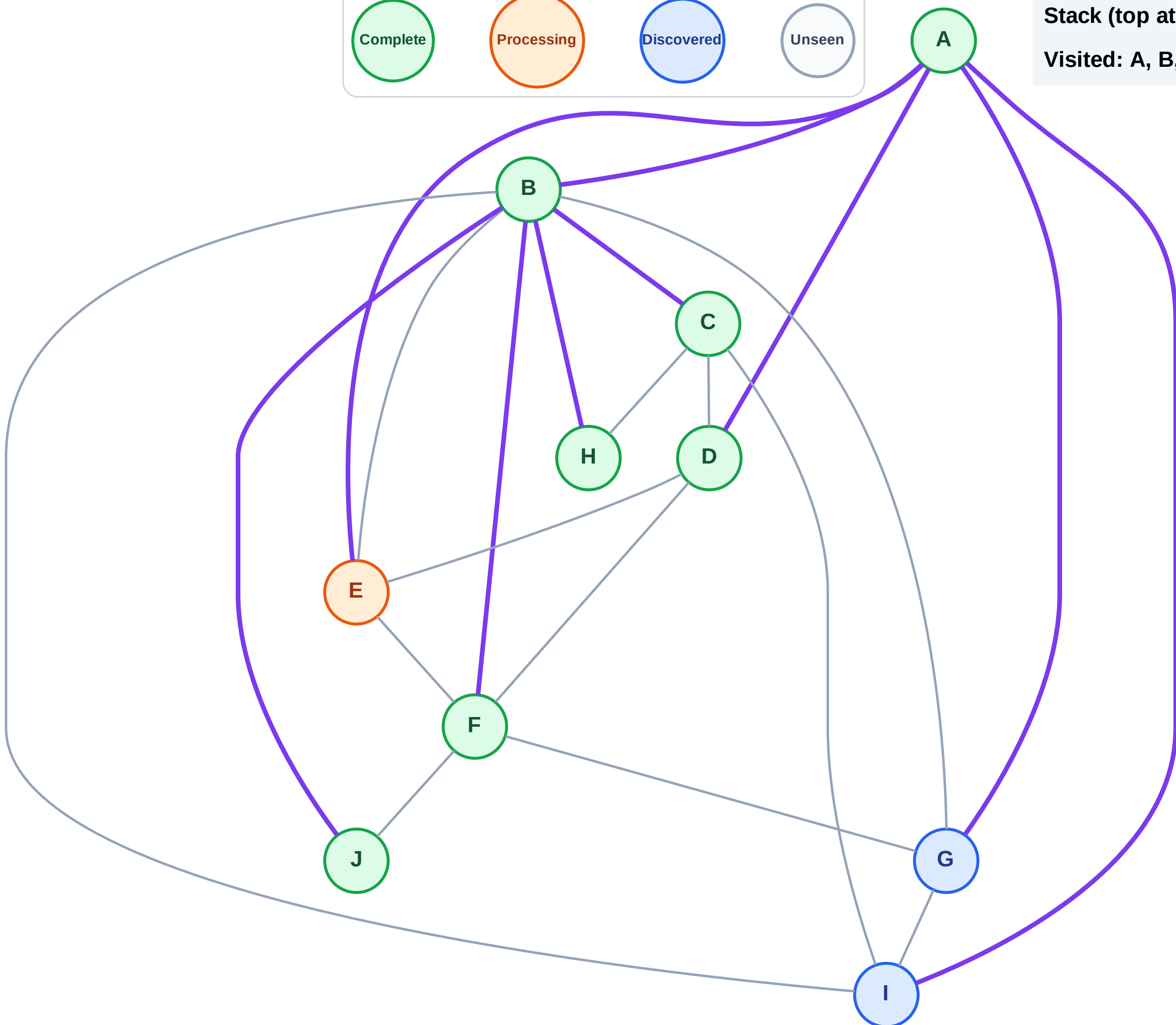
Step 16: Process node E

## Legend



Stack (top at right): I | G

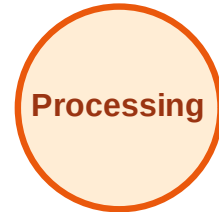
Visited: A, B, C, D, F, H, J



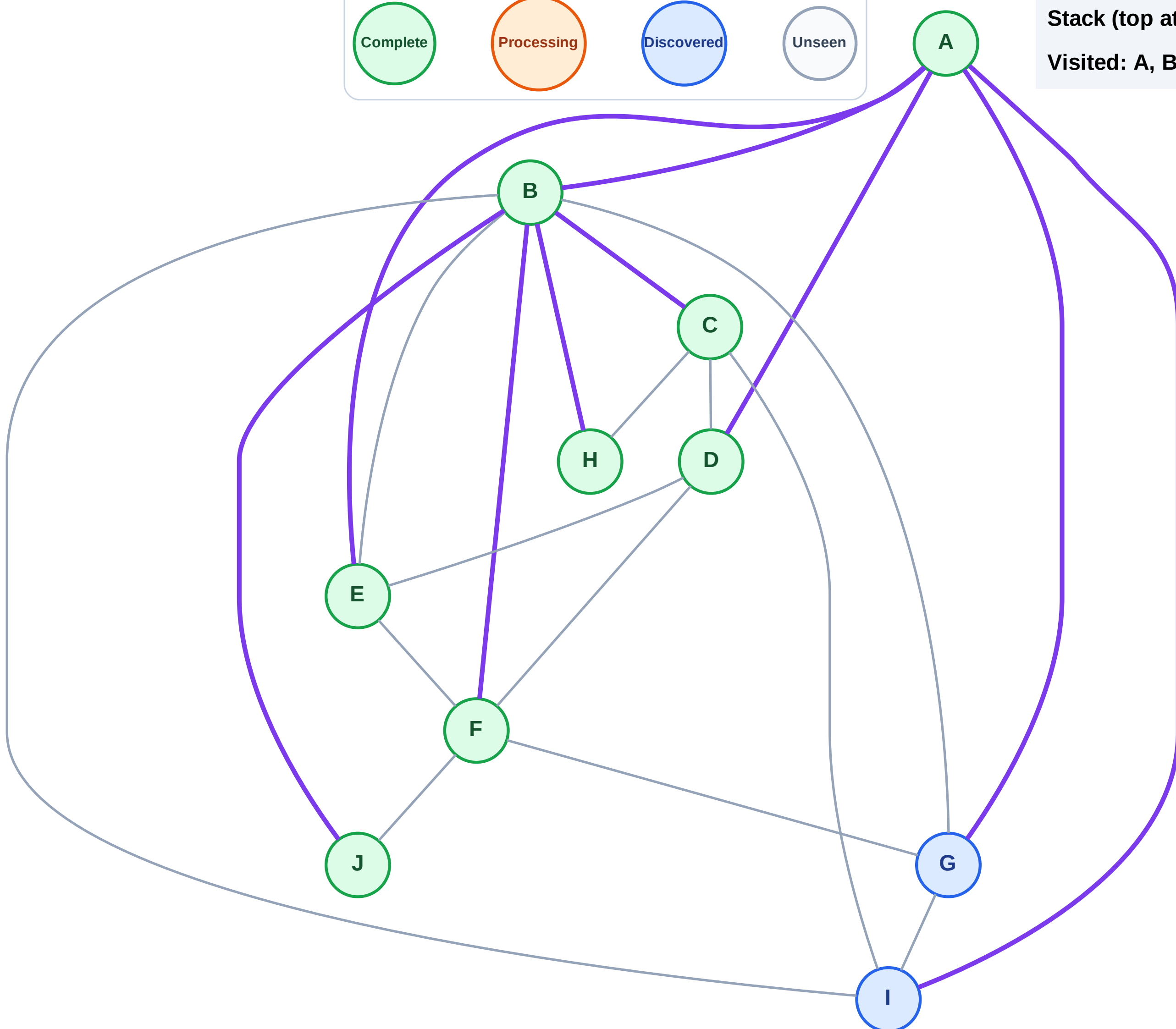
# DFS Traversal

Step 17: No new neighbors from E

## Legend



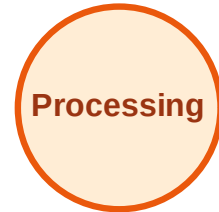
Stack (top at right): I | G  
Visited: A, B, C, D, E, F, H, J



# DFS Traversal

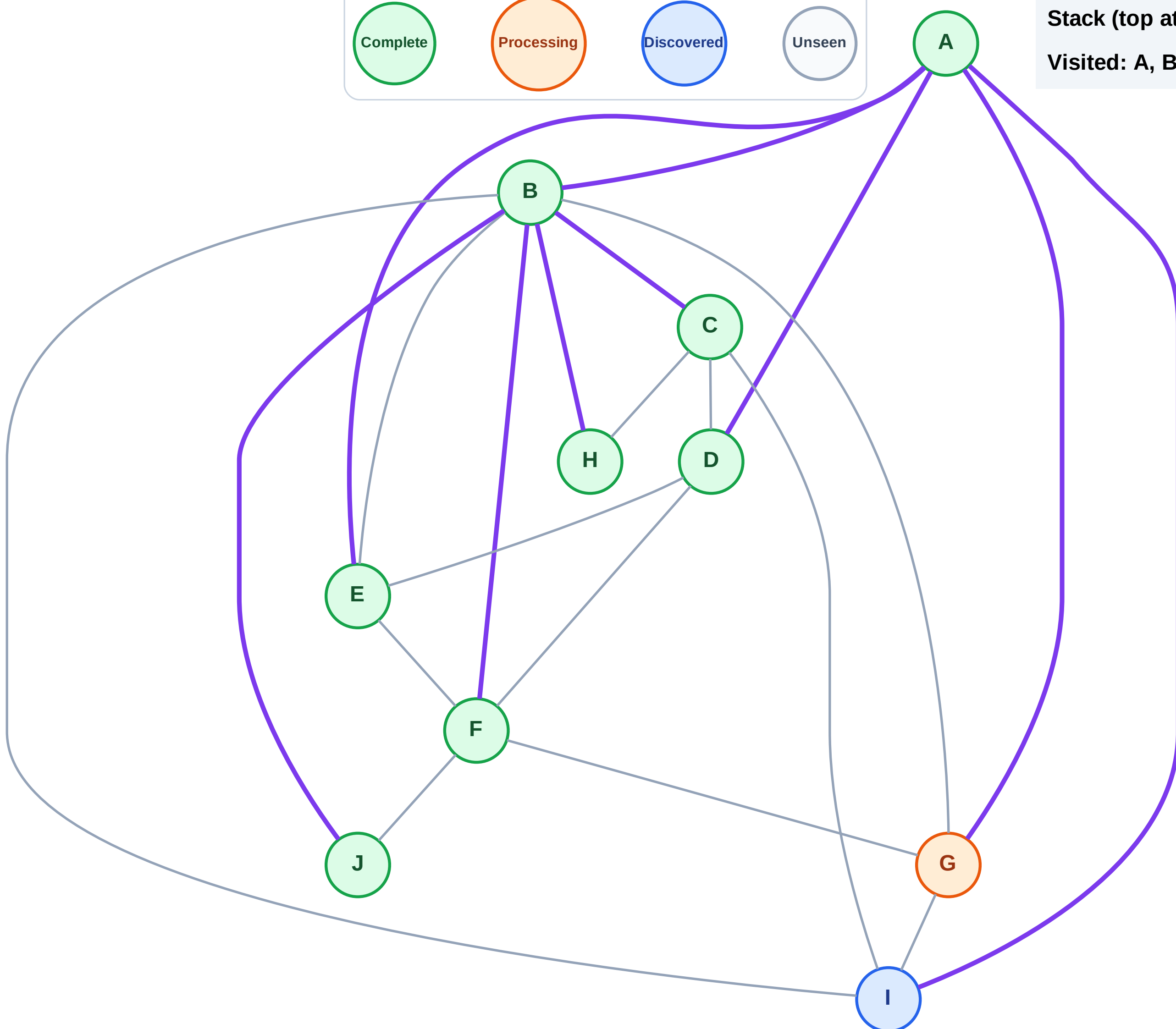
Step 18: Process node G

## Legend



Stack (top at right): I

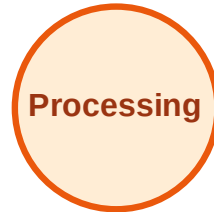
Visited: A, B, C, D, E, F, H, J



# DFS Traversal

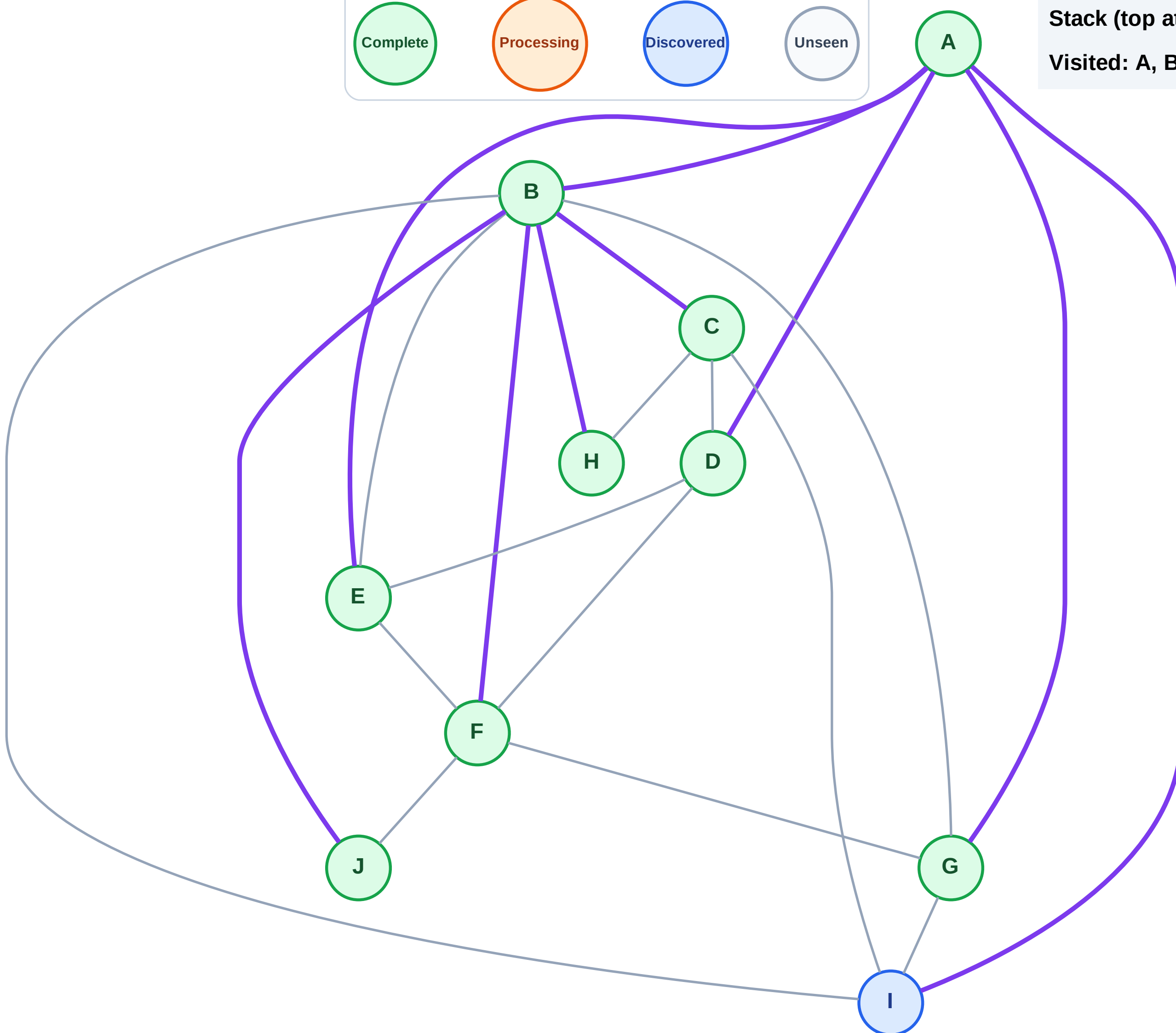
Step 19: No new neighbors from G

## Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



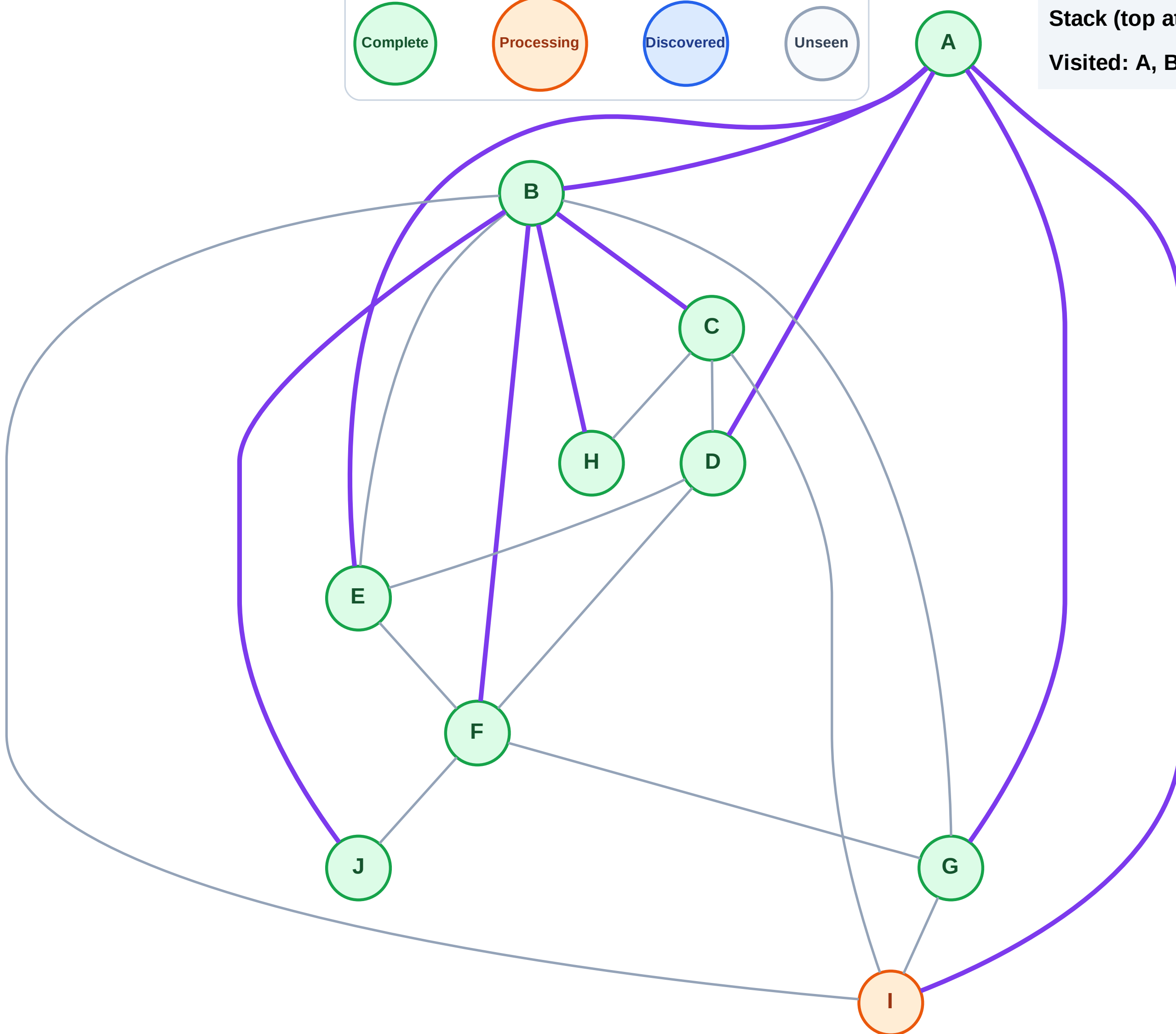
# DFS Traversal

Step 20: Process node I

## Legend



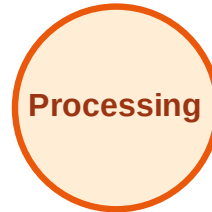
Stack (top at right): (empty)  
Visited: A, B, C, D, E, F, G, H, J



# DFS Traversal

Step 21: No new neighbors from I

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

